

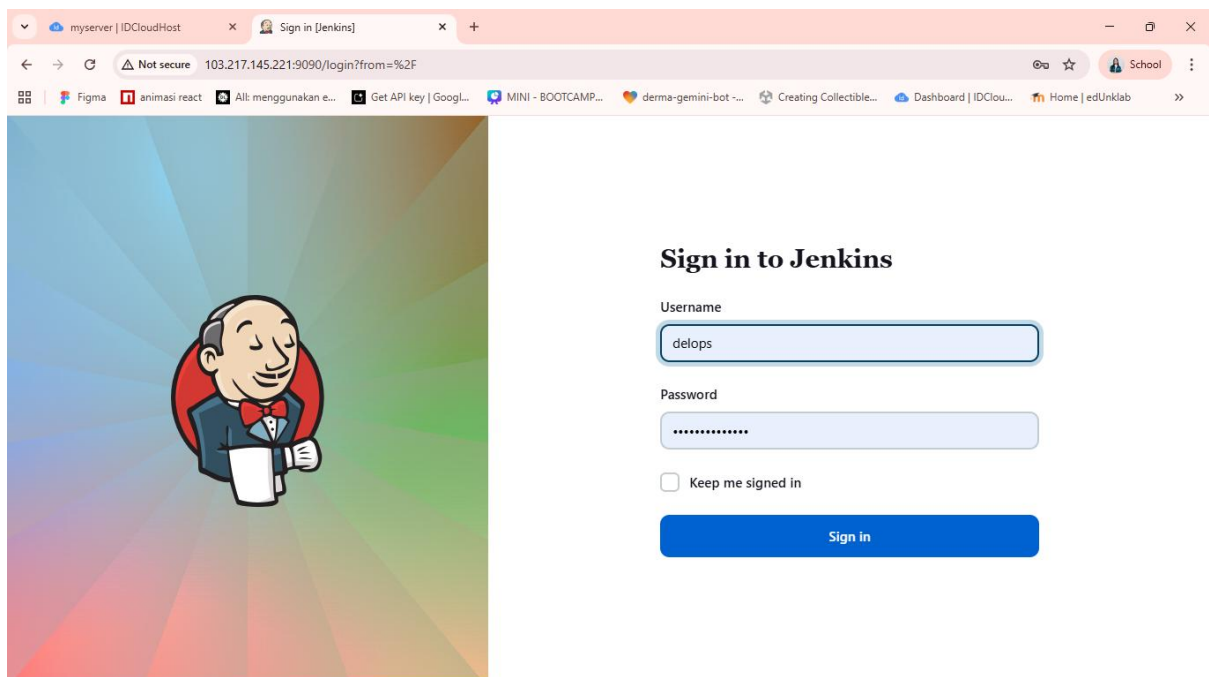
DevOps – Report CI/CD

Nama: Deeva Civia Aulia Lolong & Wendel Mnahonin

1. Login ke Server dan aktifkan Jenkins

```
root@myserver: /home/delops$ sudo su
root@myserver: /home/delops# sudo java -jar jenkins.war --httpPort=9090 &
[1] 1255
root@myserver: /home/delops# Running from: /home/delops/jenkins.war
webroot: /root/.jenkins/war
2026-04-16 03:52:17.927+0000 [id=1] INFO winstone.Logger#LogInternal: Beginning extraction from war file
2026-04-16 03:52:18.602+0000 [id=1] WARNING o.e.j.s.handler.ContextHandler#setContextPath: Empty contextPath
2026-04-16 03:52:18.728+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jetty-10.0.15; built: 2023-04-11T17:25:14.480Z; git: 68017d
bd00236bb7e187330d7585a059610f661d; jvm 11.0.30+7-post-Ubuntu-1ubuntu122.04
2026-04-16 03:52:20.059+0000 [id=1] INFO o.e.j.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jet
ty.jsp.JettyJspServlet
2026-04-16 03:52:20.264+0000 [id=1] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
root@myserver: /home/delops# 2026-04-16 03:52:21.810+0000 [id=1] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: /root/.jenkins
found at: $user.home/.jenkins
2026-04-16 03:52:22.360+0000 [id=1] INFO o.e.j.s.handler.ContextHandler#doStart: Started w.@443dbe42{Jenkins v2.414.1,,file:///root/.jenkins
/war/,AVAILABLE}{/root/.jenkins/war}
2026-04-16 03:52:22.418+0000 [id=1] INFO o.e.j.server.AbstractConnector#doStart: Started ServerConnector@23bb8443{HTTP/1.1, (http/1.1)}{0.0.0
.0:9090}
2026-04-16 03:52:22.485+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: Started Server@62e136d3{STARTING}[10.0.15,sto=0] @7906ms
2026-04-16 03:52:22.495+0000 [id=23] INFO winstone.Logger#LogInternal: Winstone Servlet Engine running: controlPort=disabled
2026-04-16 03:52:23.300+0000 [id=29] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-04-16 03:52:23.319+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-04-16 03:52:25.971+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-04-16 03:52:25.986+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-04-16 03:52:25.998+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-04-16 03:52:27.167+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-04-16 03:52:27.170+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-04-16 03:52:27.317+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-04-16 03:52:27.320+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-04-16 03:52:27.518+0000 [id=44] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
WARNING: An illegal reflective access operation has occurred
WARNING: Illegal reflective access by org.codehaus.groovy.vmplugin.v7.Java7$1 (file:/root/.jenkins/war/WEB-INF/lib/groovy-all-2.4.21.jar) to constru
ctor java.lang.invoke.MethodHandles$Lookup(java.lang.Class,int)
WARNING: Please consider reporting this to the maintainers of org.codehaus.groovy.vmplugin.v7.Java7$1
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
2026-04-16 03:52:28.294+0000 [id=28] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
```

2. Login ke dalam Jenkins



myserver | IDCloudHost x Sign in [Jenkins] x +

Not secure 103.217.145.221:9090/login?from=%2F

Sign in to Jenkins

Username

delops

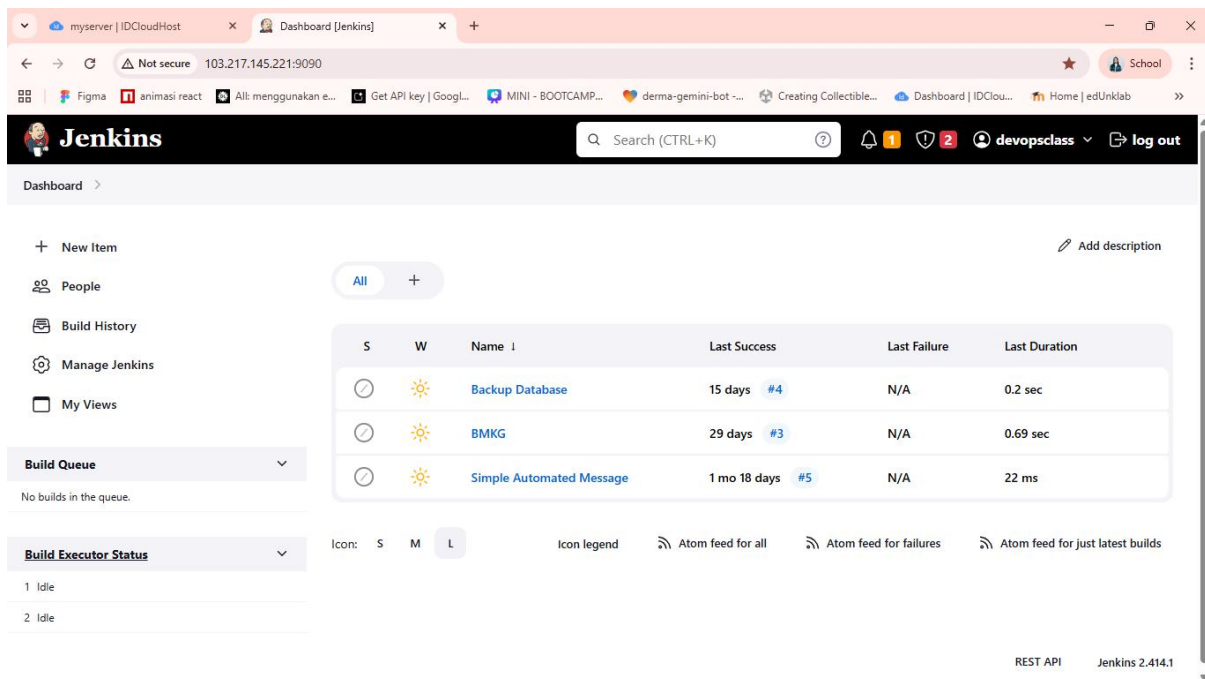
Password

.....

☐ Keep me signed in

Sign in

Tampilan Dashboard Jenkins

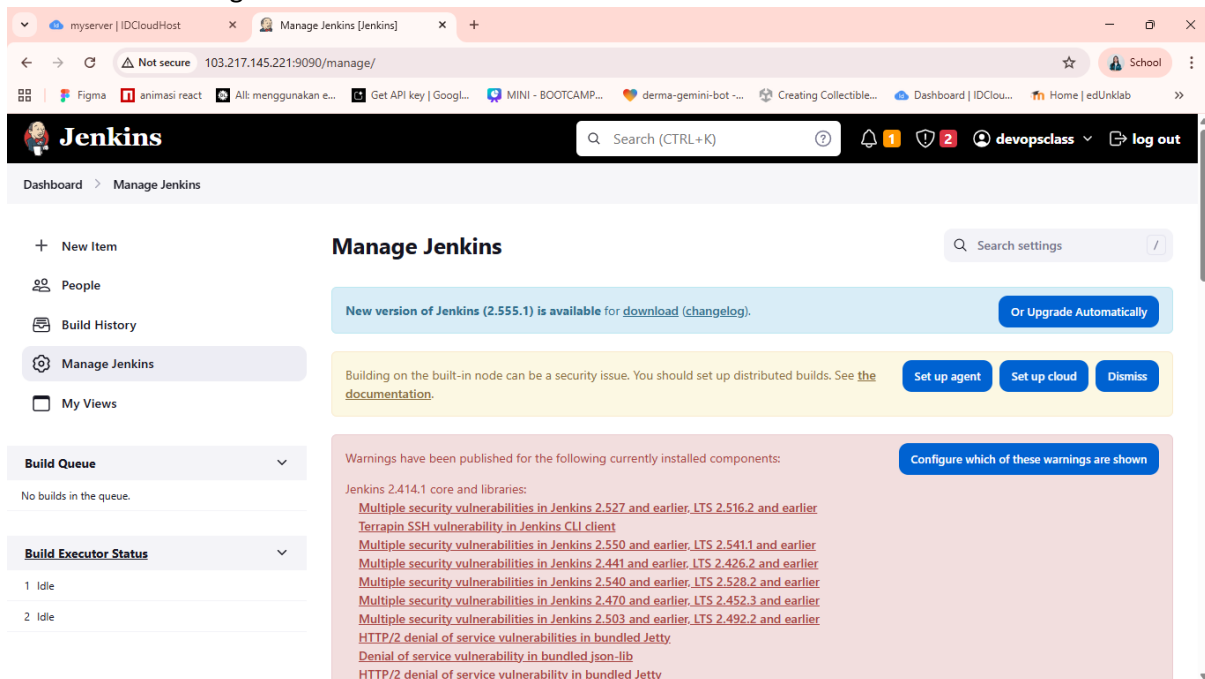


The screenshot shows the Jenkins Dashboard in a web browser. The top navigation bar includes the Jenkins logo, a search bar, and user information. The left sidebar contains links to 'New Item', 'People', 'Build History', 'Manage Jenkins', and 'My Views'. The main content area displays a table of builds with columns for status, name, last success, last failure, and last duration. Below the table, there are sections for 'Build Queue' and 'Build Executor Status'.

S	W	Name	Last Success	Last Failure	Last Duration
🕒	☀️	Backup Database	15 days #4	N/A	0.2 sec
🕒	☀️	BMKG	29 days #3	N/A	0.69 sec
🕒	☀️	Simple Automated Message	1 mo 18 days #5	N/A	22 ms

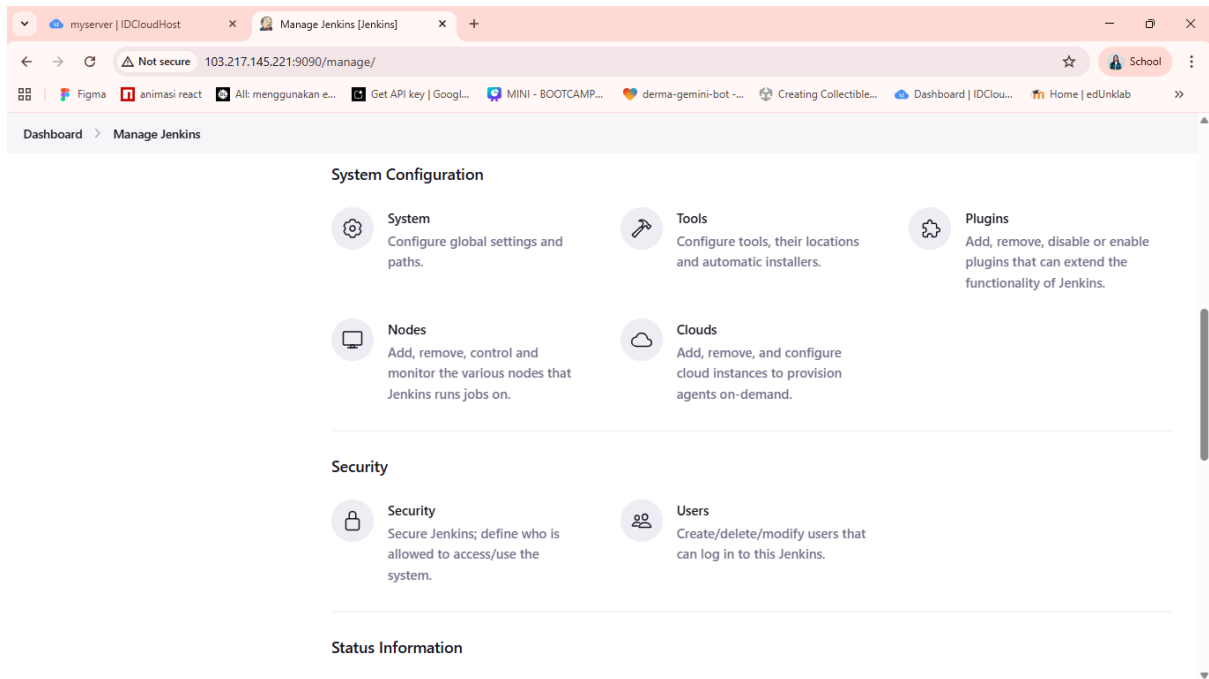
3. Install Git Plugin

Buka menu “Manage Jenkins”

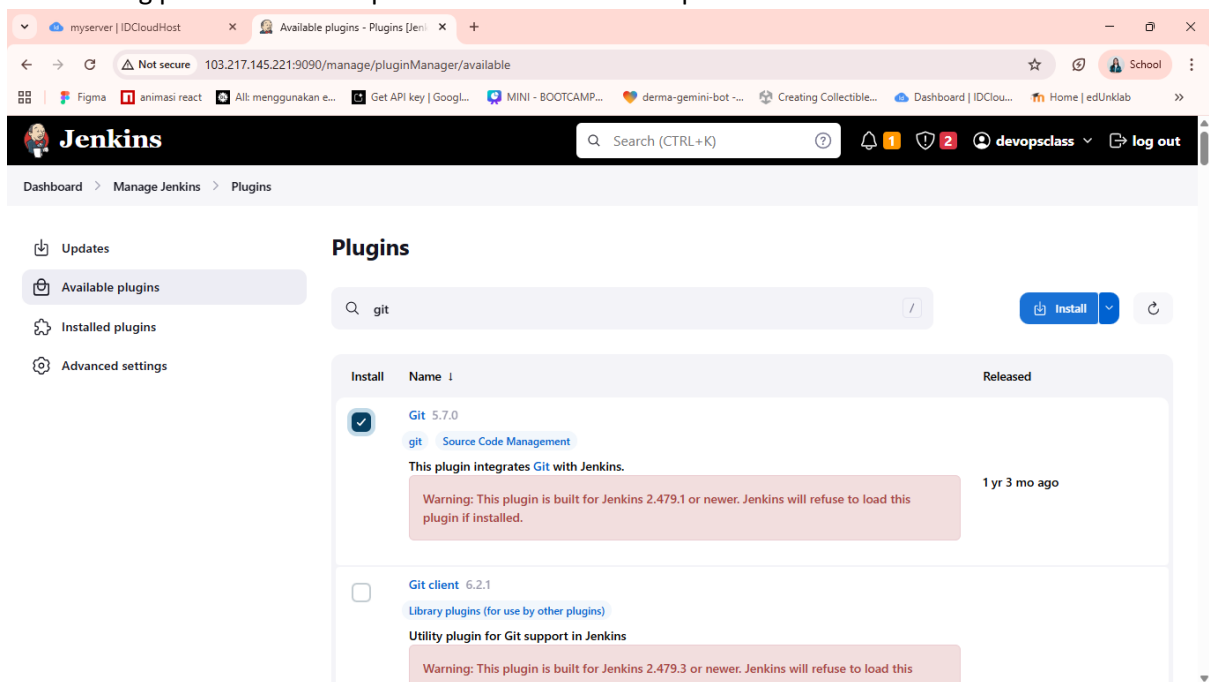


The screenshot shows the 'Manage Jenkins' page in the Jenkins web interface. The top navigation bar is the same as the dashboard. The left sidebar is also the same. The main content area is titled 'Manage Jenkins' and includes a search bar for settings. Below the title, there are several sections: a notification about a new version of Jenkins (2.555.1), a warning about building on the built-in node, and a section for warnings published for currently installed components. The 'Warnings' section lists several security vulnerabilities in Jenkins and its dependencies, including Jenkins 2.414.1 core and libraries, and various security vulnerabilities in Jenkins CLI client, Jenkins 2.550 and earlier, Jenkins 2.441 and earlier, Jenkins 2.540 and earlier, Jenkins 2.470 and earlier, Jenkins 2.503 and earlier, and HTTP/2 denial of service vulnerabilities in bundled Jetty.

Pilih menu “Plugins”



Pilih “Available plugins” pada sidebar dan cari “git”.
lalu centang pada row Git dan pilih “install after restart” pada tombol install.



myserver | IDCLOUDHost x Download progress - Plugins | x +

Not secure 103.217.145.221:9090/manage/pluginManager/updates/ School

Figma animasi react All: menggunakan e... Get API key | Googl... MINI - BOOTCAMP... derma-gemini-bot ... Creating Collectible... Dashboard | IDClou... Home | edUnklab

Jenkins Search (CTRL+K) devopsclass log out

Dashboard > Manage Jenkins > Plugins

Download progress

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

Structs

Pipeline: Step API

commons-lang3 v3.x Jenkins API

commons-text API

bouncycastle API

Credentials

Plain Credentials

Variant

SSH Credentials

Credentials Binding

Pipeline: SCM Step

Mina SSHD API :: Common

Mina SSHD API :: Core

Downloaded Successfully. Will be activated during the next boot

Downloaded Successfully. Will be activated during the next boot

Downloaded Successfully. Will be activated during the next boot

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Downloaded Successfully. Will be activated during the next boot

Downloaded Successfully. Will be activated during the next boot

Setelah selesai download, scroll ke paling bawah dan centang pilihan “Restart Jenkins when installation is complete and no jobs are running”.

lalu tampilan akan menjadi seperti ini:

myserver | IDCLOUDHost x Restarting Jenkins x +

Not secure 103.217.145.221:9090/manage/pluginManager/updates/ School

Figma animasi react All: menggunakan e... Get API key | Googl... MINI - BOOTCAMP... derma-gemini-bot ... Creating Collectible... Dashboard | IDClou... Home | edUnklab

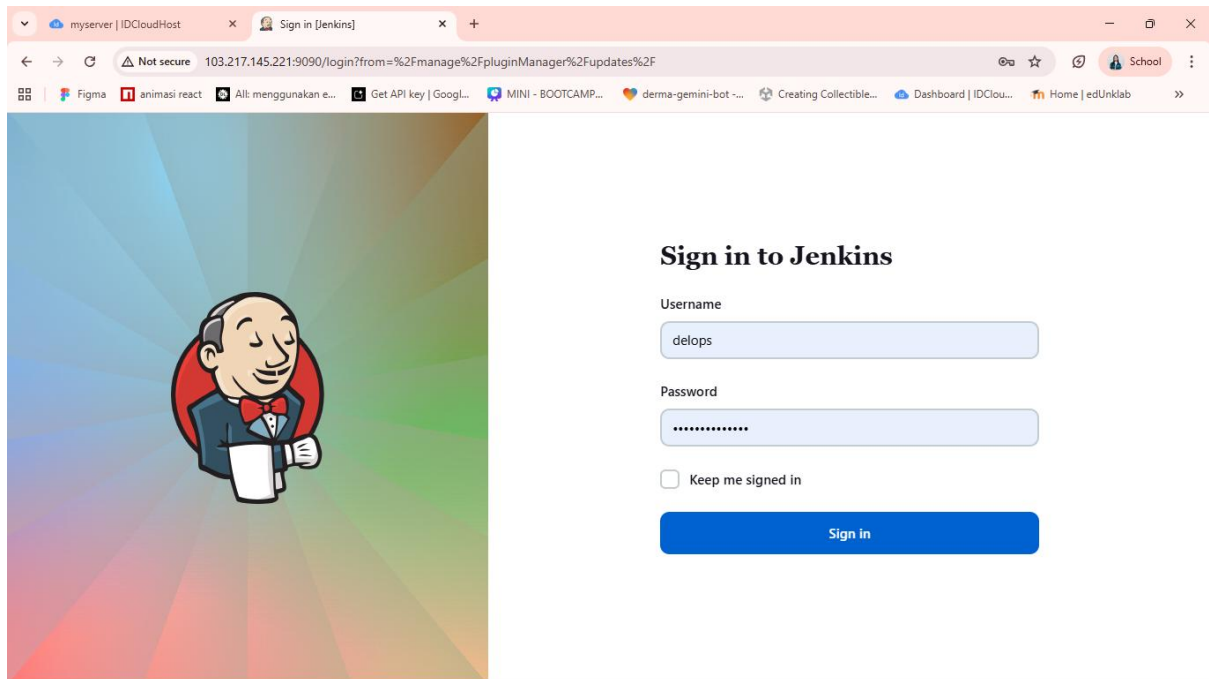


Please wait while Jenkins is restarting ...

Your browser will reload automatically when Jenkins is ready.

Safe Restart
Builds on agents can usually continue.

Setelah selesai restart, login lagi ke dalam jenkins



Note: kalau tampilan git di dalam “installed plugins” masih merah, berarti harus update jenkins war ke versi terbaru, lalu update juga java ke versi terbaru. Setelah itu jalankan service jenkins lagi.

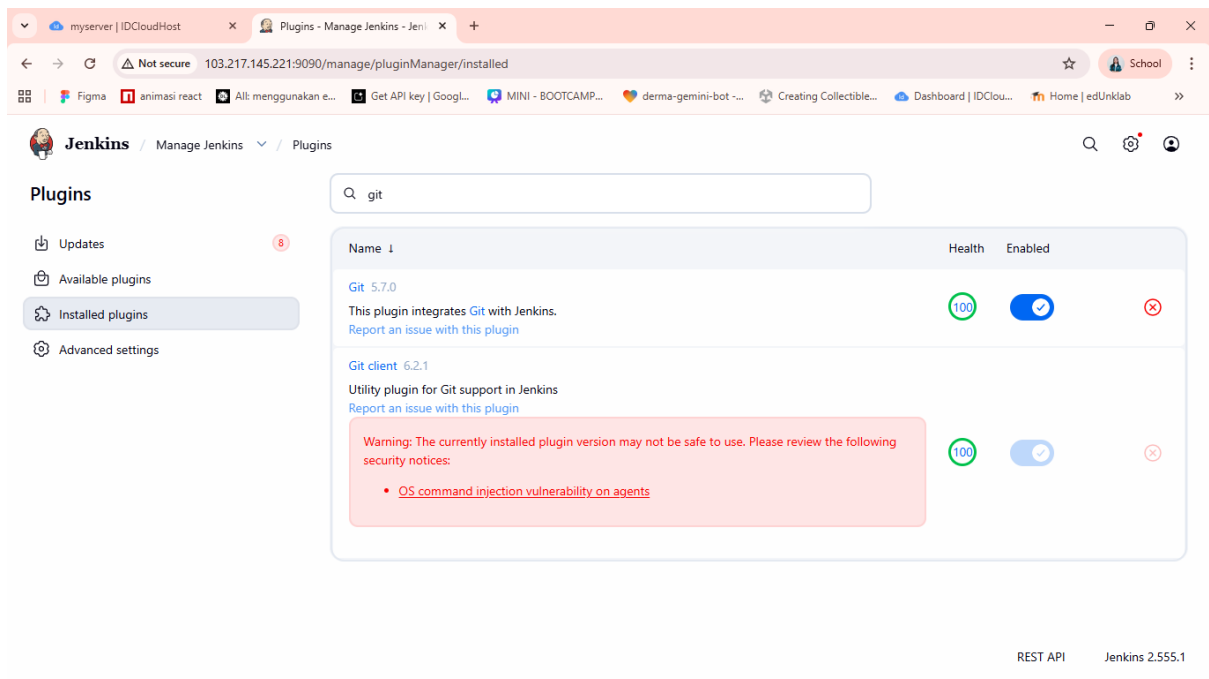
Code Update jenkins: `wget https://get.jenkins.io/war-stable/latest/jenkins.war -O jenkins.war`

Code Update Java:

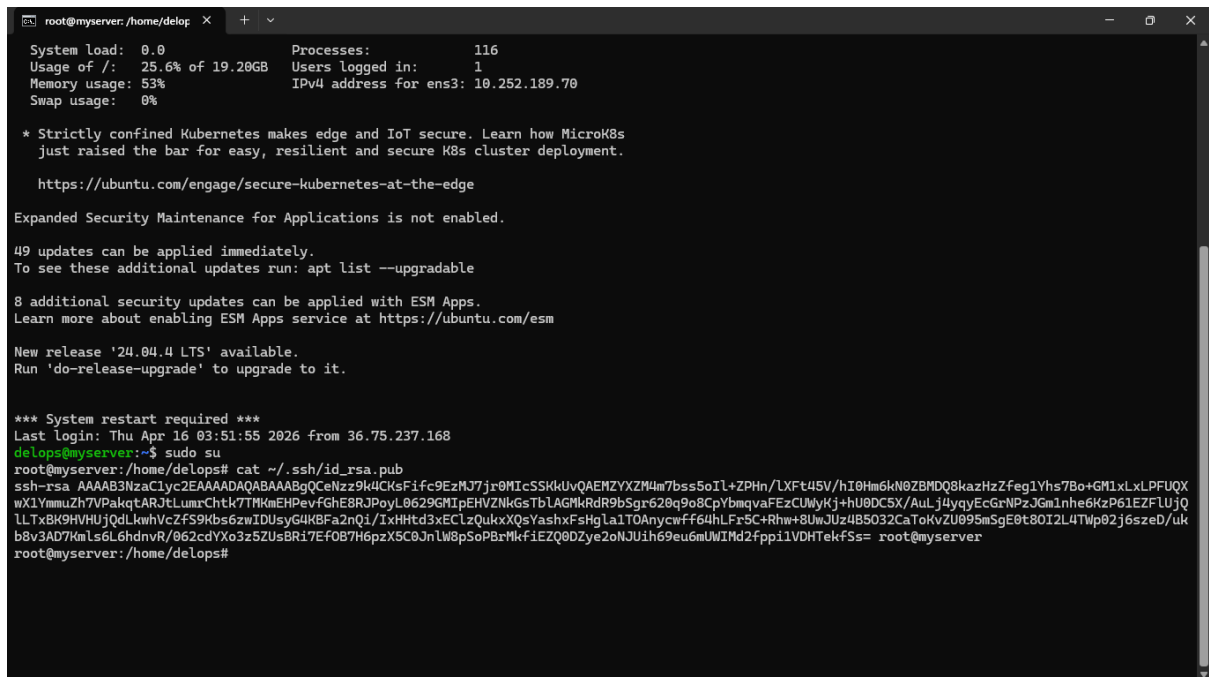
- `sudo apt update`
- `sudo apt install fontconfig openjdk-21-jre -y`
- `sudo update-alternatives --config java` → ini untuk ubah default java ke versi terbaru (pilih versi 21)
- `java -version` → cek versi Java yang sekarang

Jalankan kembali service jenkins: `sudo java -jar jenkins.war --httpPort=9090 &`

Maka tampilan dalam “installed plugins” akan jadi seperti ini:



4. Cek Public Key



Copy public key ini, lalu paste ke dalam github

```
root@myserver: /home/delops x + v
System load: 0.0 Processes: 116
Usage of /: 25.6% of 19.20GB Users logged in: 1
Memory usage: 53% IPv4 address for ens3: 10.252.189.70
Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

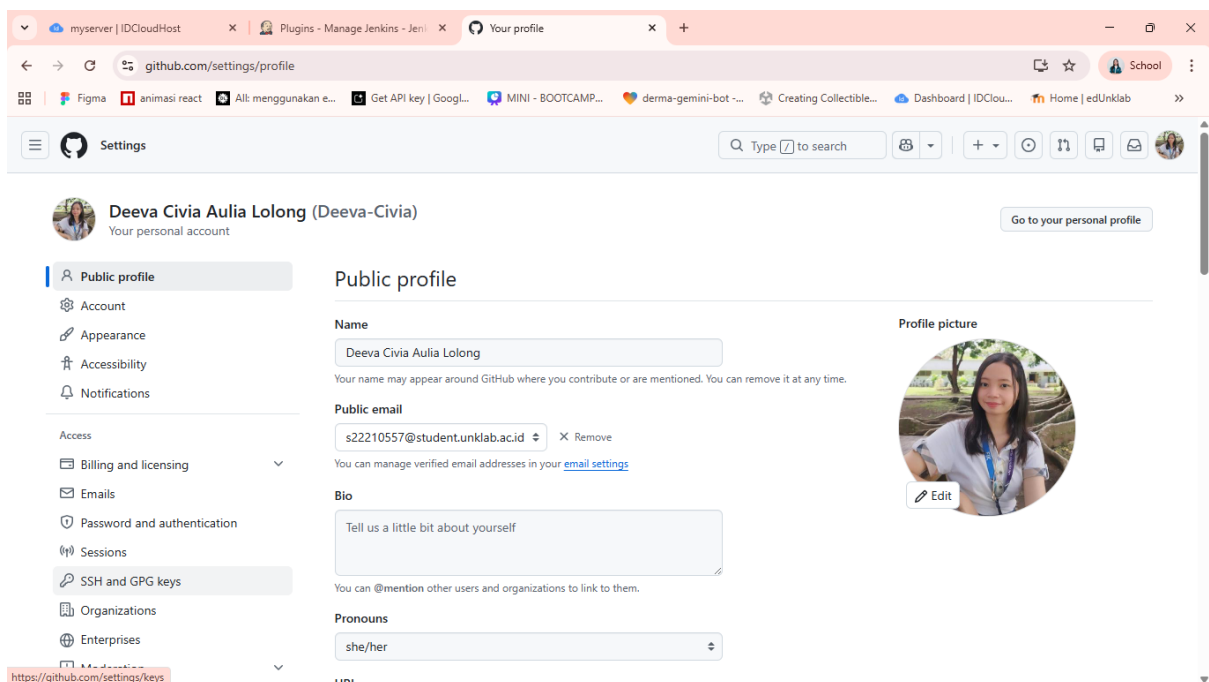
49 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

8 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

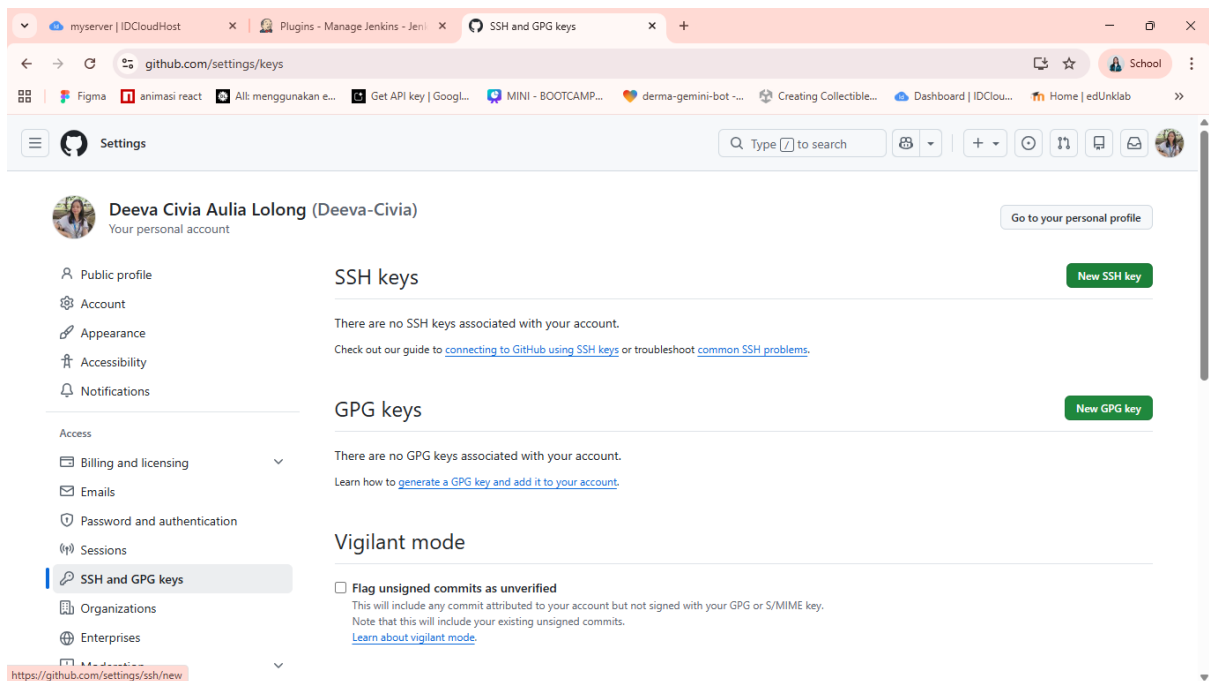
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

*** System restart required ***
Last login: Thu Apr 16 03:51:55 2026 from 36.75.237.168
delops@myserver:~$ sudo su
root@myserver: /home/delops# cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCEAz9k4CKsFifc9EzMJ7jr0MICSSkKuvQAEMZYXZM4m7bss5oIl+ZPhn/LXft45V/hI0Hm6kN0ZBMDQ8kazHzZfeg1Yhs7Bo+GM1xLxLPFUQX
wX1YmmuZ7V7PakqtARJtLumrChtk7TMMkEHPevfGhE8RJPoyL0629GMIpEHVZNIkGsTbLAGMkRdR9bSgr620q9o8CpYbmqvaFEzCUWYKj+hU0DC5X/AuLj4yqyEcGrNPzJGmInhe6KzP61EZFLUjQ
LLTx8K9HVHujQdLkwhVcZfS9Kbs6zwIDUsyG4KBFa2nQi/IxHHTd3xECIzQukxXQsYashxFsHgla1TOAnycwff64hLFr5C+Rhw+8UwJUz4B5032CaToKvZU095mSgE0t8OI2L4Twp02j6szeD/uk
b8v3AD7KmlS6L6hdnVR/062cdYXo3z5ZUsBRi7EfoB7H6pzX5C0JnLw8pSoPBkMkfiEZQ0DZye2oNJuIh69eu6mUwIMd2fppi1VDHTekfSs= root@myserver
root@myserver: /home/delops#
```

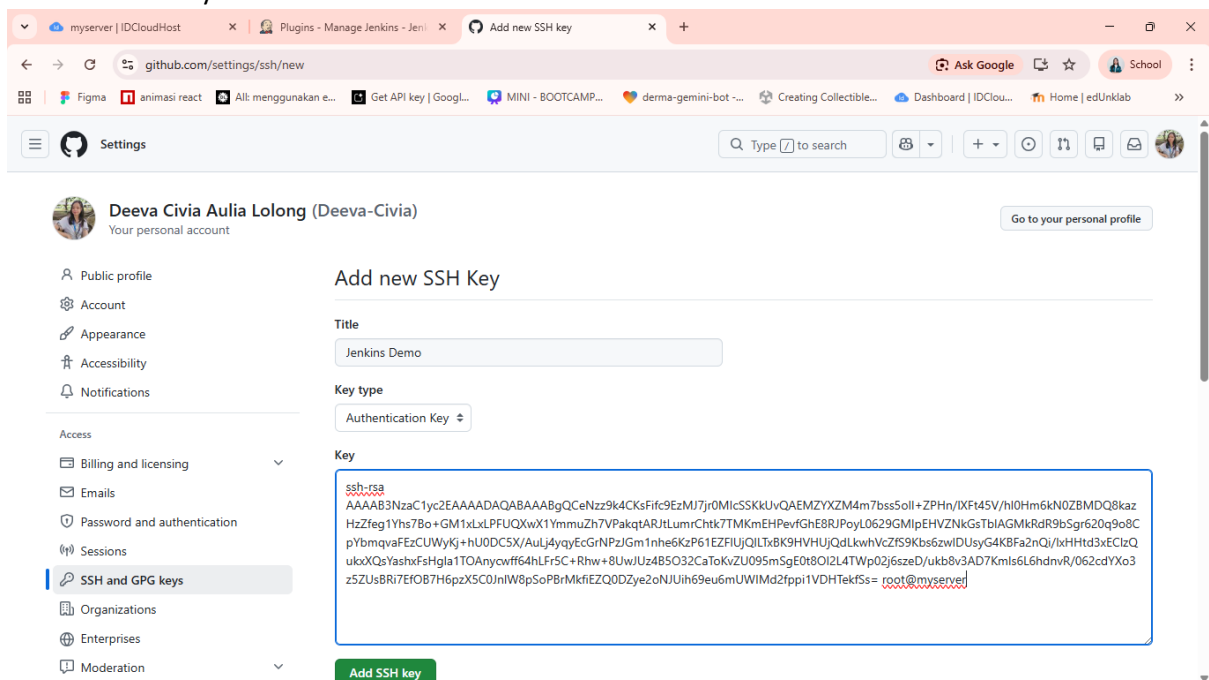
Buka Settings di profile Github



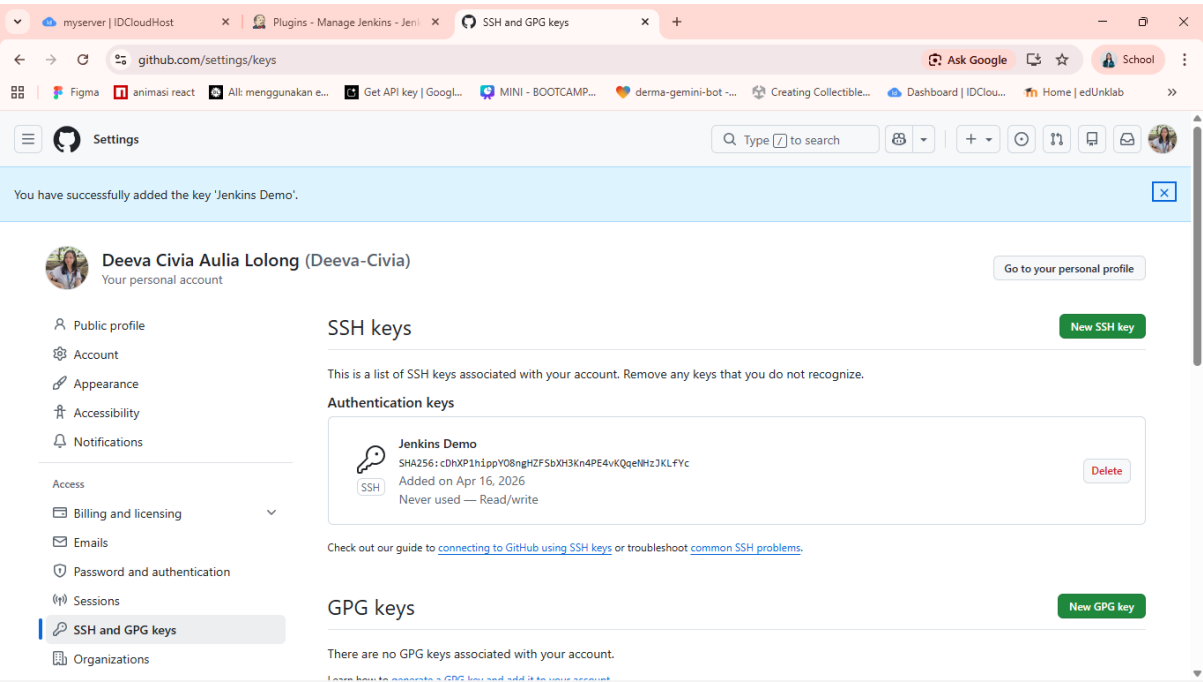
Pilih menu “SSH and GPG keys”. Lalu pilih “New SSH key”



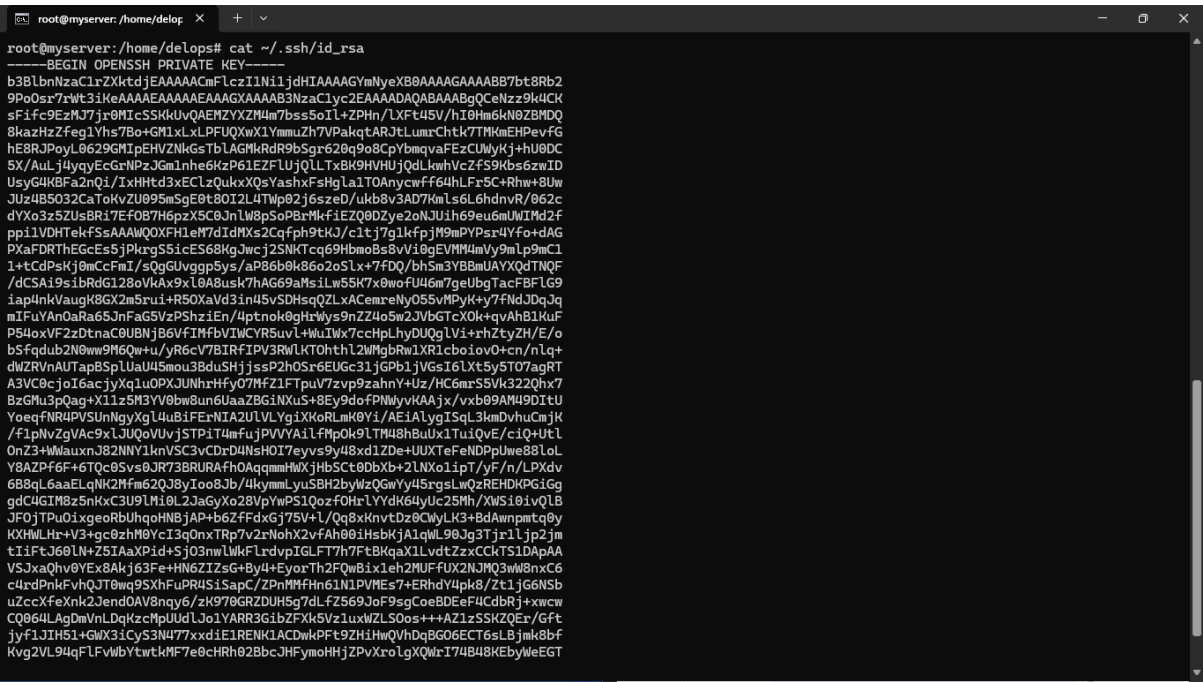
Masukkan public key yang telah di salin tadi ke dalam kolom “Key”, beri Title “Jenkins Demo”, lalu klik “Add SSH key”.



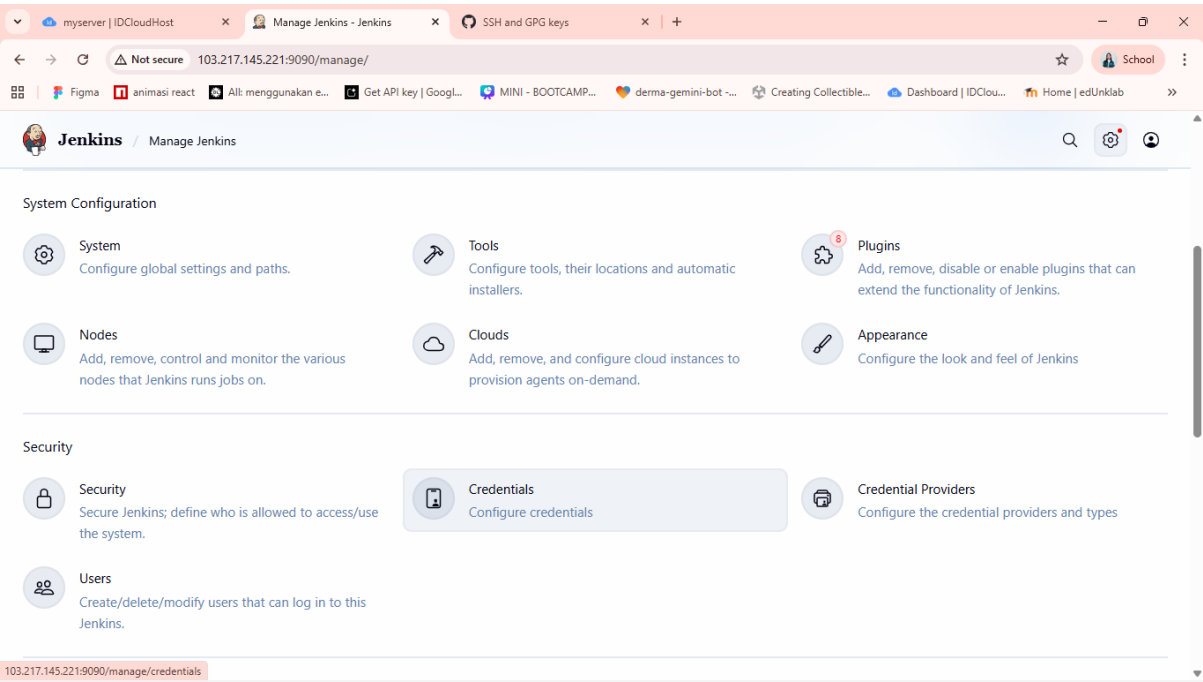
Kalau sudah seperti ini, berarti telah berhasil menyambungkan server dengan github melalui authentication keys.



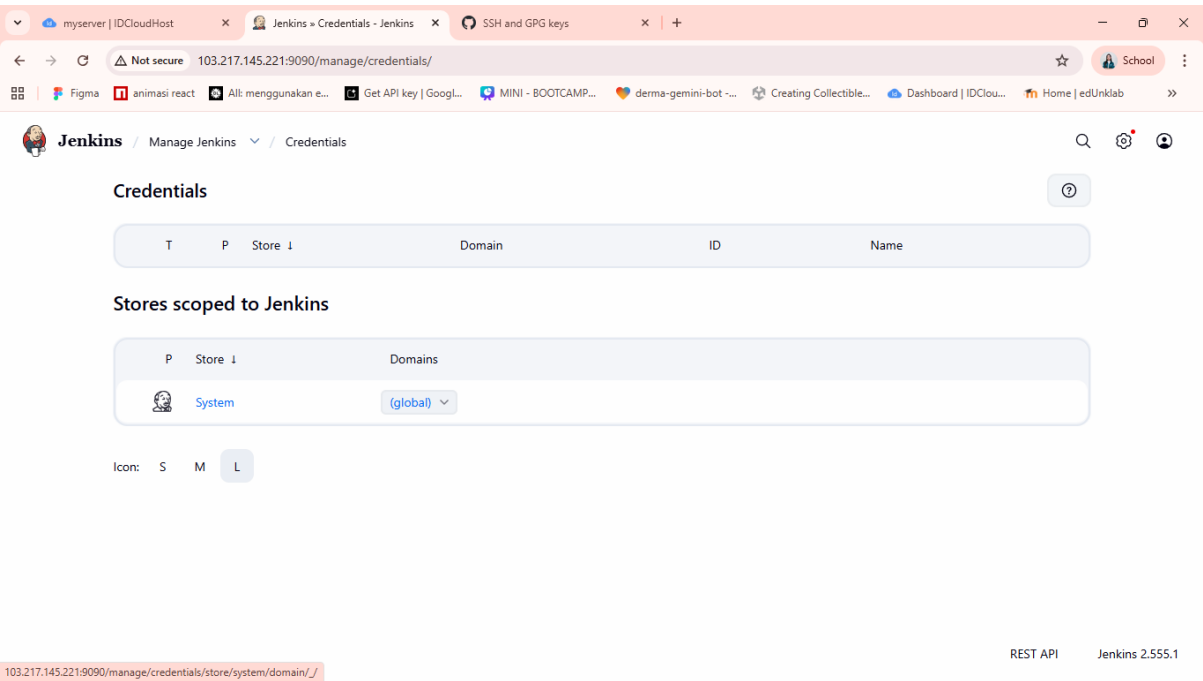
Sekarang kita ambil Private Key



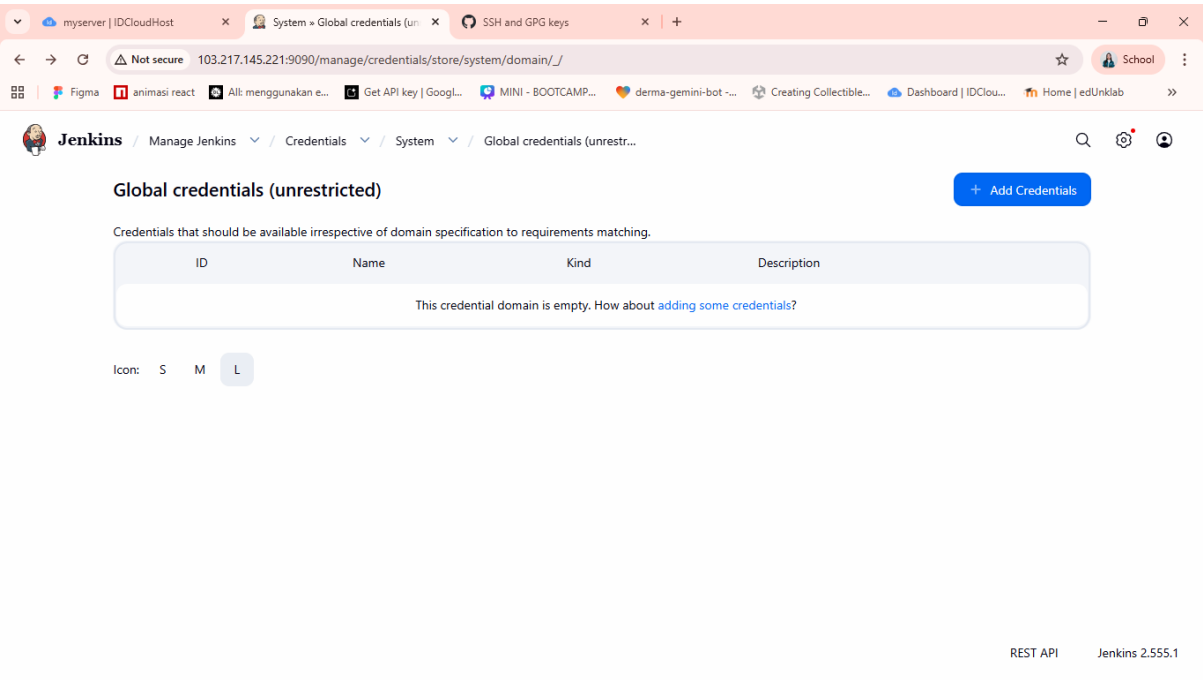
Buka Jenkins, dan buka menu “Menage Jenkins”, lalu pilih “Credentials”



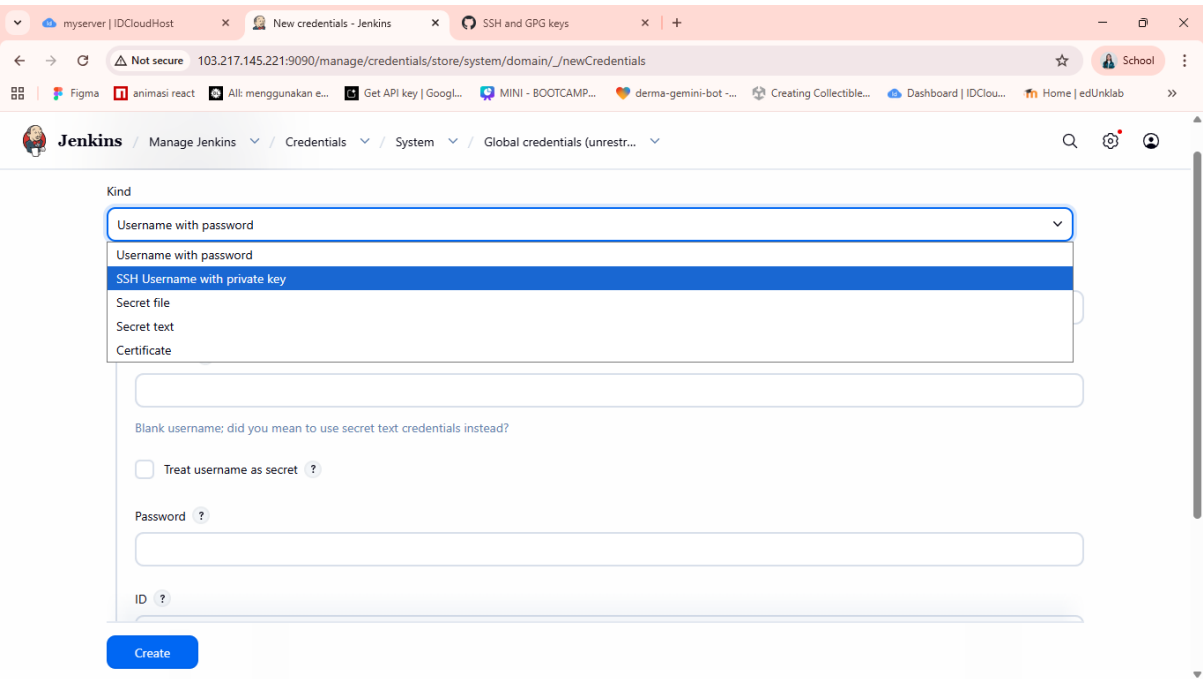
Pilih global, untuk membuat global credentials



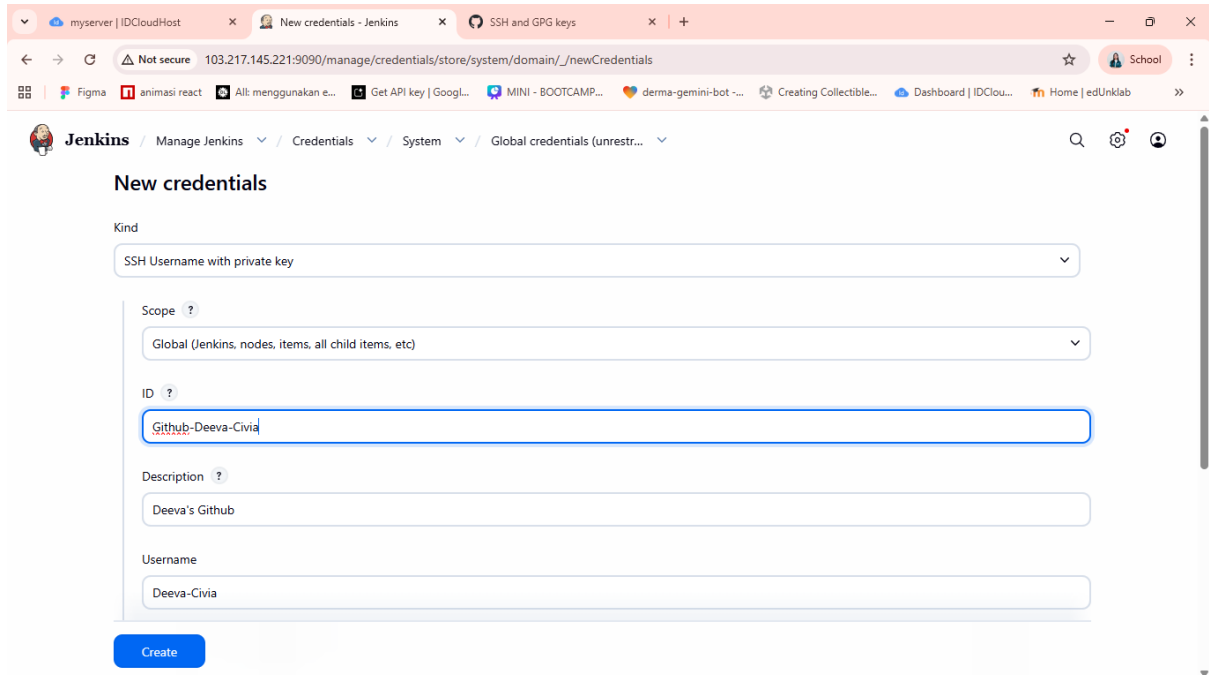
Klik “Add Credentials”



Dalam field “Kind”, pilih “SSH username with private key”



Masukkan username, untuk ID dan Description itu Optional.



The screenshot shows the Jenkins 'New credentials' page. The 'Kind' dropdown is set to 'SSH Username with private key'. The 'Scope' dropdown is set to 'Global (Jenkins, nodes, items, all child items, etc)'. The 'ID' field contains 'Github-Deeva-Civia'. The 'Description' field contains 'Deeva's Github'. The 'Username' field contains 'Deeva-Civia'. A blue 'Create' button is at the bottom.

Kind

SSH Username with private key

Scope ?

Global (Jenkins, nodes, items, all child items, etc)

ID ?

Github-Deeva-Civia

Description ?

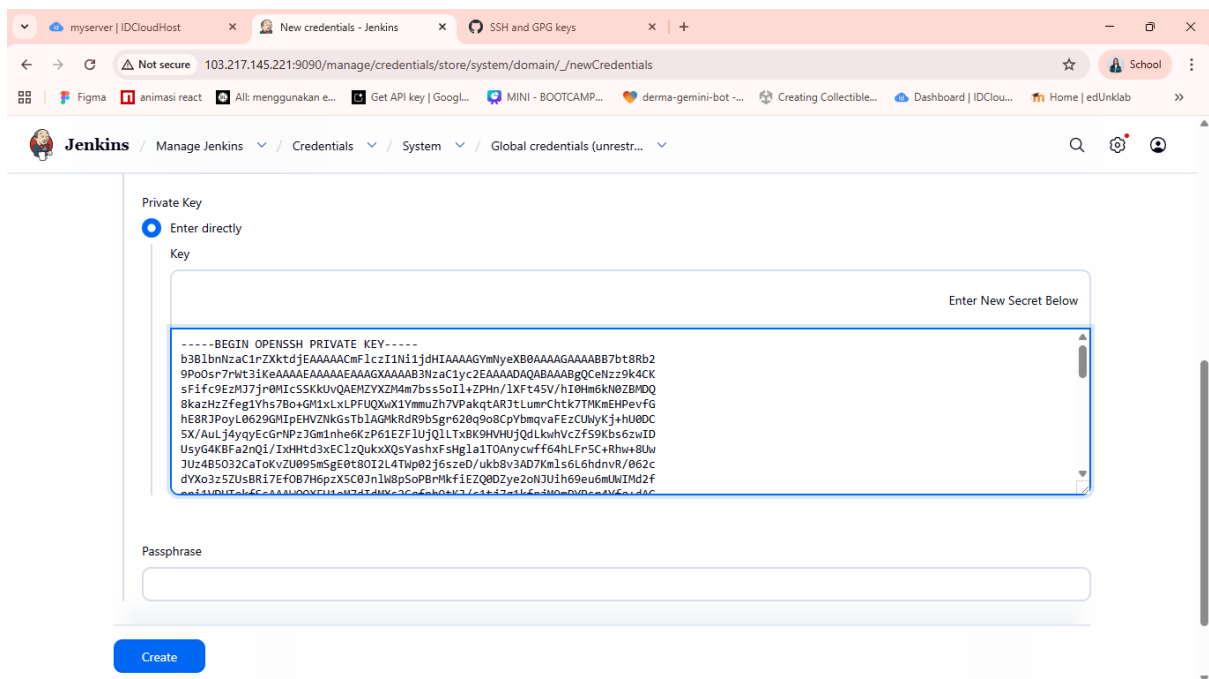
Deeva's Github

Username

Deeva-Civia

Create

Lalu pilih "Enter directly", dan masukkan private key yang telah diambil dari server tadi. Dan klik Create



The screenshot shows the Jenkins 'New credentials' page with the 'Private Key' section expanded. The 'Enter directly' radio button is selected. The 'Key' field contains a long SSH private key. The 'Passphrase' field is empty. A blue 'Create' button is at the bottom.

Private Key

Enter directly

Key

Enter New Secret Below

-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnRzaC1rZXktdjEAAAAAAAAAAAAAIA1N11jdHIAAAAGYmlyeXB0AAAAAAB7b28Rb2
9Po0sr7rWt31keAAAAEAAAAEAAAGXAAAB3NzaC1yc2EAAAADAQABAAQGCeNzz9k4CK
sFifc9EzMJ7jrbMIcSSKkUvQAEHYZXZM4m7bss5o1L+ZPHn/1Xft45V/h10Hm6kN0ZBMDQ
8kazHz2feg1Yhs7Bo+GM1xLxLPUQUXwX1YmmuZ7h7VPakgtAR3tLumChkt7TMkNEHPvfg
hEBRJPoyL0629GMIpEHVZnKgsTb1AGMkRdr9bSgr620q908CpYbmqvaFeZCUmyKj+hU8DC
5X/Aul-j4yqyEcGrNPz3Gm1nhe6KzP61EZf1UjQ1LTxBK9HVHujQdLkwhVcZf59Kbs6zwID
UsyG4KBFa2nQ1/IxHtd3xEC1zQukxxQsYashxFsHg1a1TOAnycwff64hLFr5C+Rhw+8Uw
JUz4B5032CaTokvZU095mSgE0t80I2L4Twp02j6szed/ukb8v3AD7Km1s6L6hdnvr/062c
dYXo3z5ZUsBR17EFOB7H6pzX5C03n1W8p5oPBmKfIEZQ80Zye2oN3Uih69eu6mUWIMd2f
eei4MDUtel5CaAMUQ9VFUteM3d7dMXc3Ca6eb8k7/c1t474k6s3M0eRVDce4Y6vd46

Passphrase

Create

Ini tampilan credentials telah berhasil dibuat:

Global credentials (unrestricted)

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description
Github-Deeva-Civia	Deeva-Civia (Deeva's Github)	SSH Username with private key	Deeva's Github

Icon: S M L

REST API Jenkins 2.555.1

5. Integrasi Git dan Jenkins

Buka project yang sudah dimasukkan ke github, lalu copy SSH link yang ada:

Deeva-Civia / Registration-MIS-DevOps

main 1 Branch 0 Tags

Go to file Add file Code

Deeva-Civia first commit

- app first commit
- bootstrap first commit
- config first commit
- database first commit
- public first commit
- resources first commit
- routes first commit 2 days ago
- storage first commit 2 days ago
- tests first commit 2 days ago

Local Codespaces

Clone

HTTPS SSH GitHub CLI

git@github.com:Deeva-Civia/Registration-MIS-DevOps

Use a password-protected SSH key.

Open with GitHub Desktop

Open with Visual Studio

Download ZIP

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Buat job baru di jenkins, dengan nama “Connected to First Github”:

myserver | IDCloudHostNew Item - JenkinsDeeva-Civia/Registration-MIS-1

Not secure103.217.145.221:9090/view/all/newJob

Figmaanimasi reactAll: menggunakan e...Get API key | Googl...MINI - BOOTCAMP...derma-gemini-bot - ...Creating Collectible...Dashboard | IDClou...Home | edUnklab

JenkinsAllNew Item

New Item

Enter an item name

Connected to First Github

Select an item type

Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

Duplicate an existing item

Create a new item by copying the configuration of an existing one.

OK

REST APIJenkins 2.555.1

Masukkan Description:

myserver | IDCloudHostConnected to First Github Confideeva-Civia/Registration-MIS-1

Not secure103.217.145.221:9090/job/Connected%20to%20First%20Github/configure

Figmaanimasi reactAll: menggunakan e...Get API key | Googl...MINI - BOOTCAMP...derma-gemini-bot - ...Creating Collectible...Dashboard | IDClou...Home | edUnklab

JenkinsConnected to First GithubConfigure

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

General

Enabled

Description

Connect to github repository

Plain textPreview

☐ Discard old builds ?

☐ This project is parameterized ?

☐ Execute concurrent builds if necessary ?

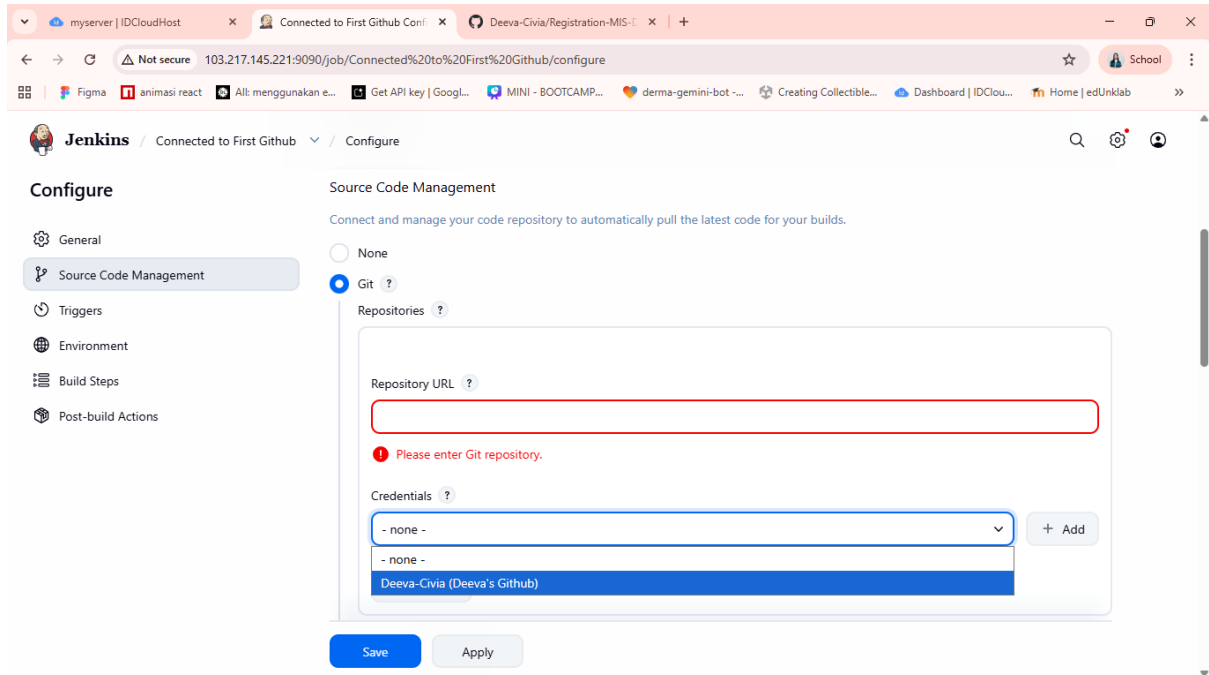
Advanced

Source Code Management

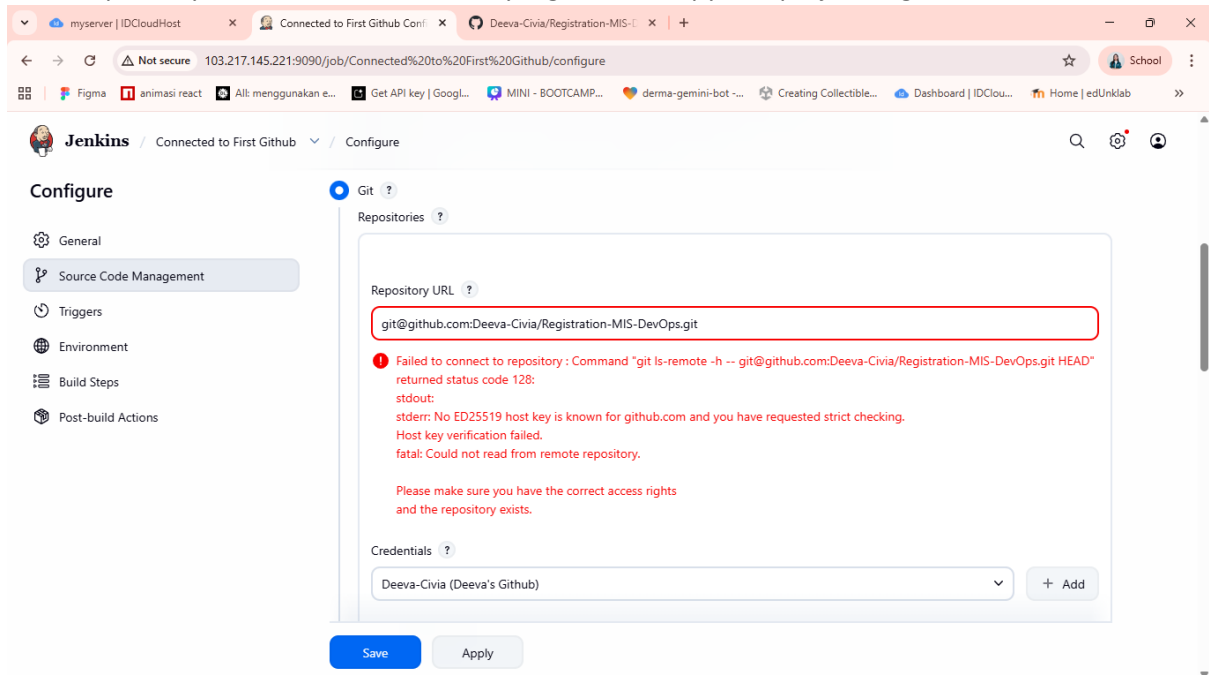
Save

Apply

Pada bagian “Source Code Management” pilih credentials yang telah dibuat tadi:



Lalu “Repository URL” masukkan SSH link yang telah di copy dari project di github.



Kalau muncul failed seperti ini, berarti server Jenkins belum mengenali atau belum percaya pada identitas server GitHub (github.com). Hal ini karena secara bawaan, protokol SSH menerapkan pengecekan ketat. Ketika Jenkins mencoba menghubungi GitHub untuk pertama kalinya, GitHub akan mengirimkan *host key*. Karena kunci ini belum ada di dalam daftar *known_hosts* di server, dan Jenkins berjalan di latar belakang, koneksi tersebut otomatis ditolak oleh sistem.

Maka kita harus menjalankan perintah ini di dalam server:

- `mkdir -p ~/.ssh` → Membuat folder `.ssh` di server. Folder ini untuk menyimpan semua konfigurasi dan kunci keamanan.
- `ssh-keyscan github.com >> ~/.ssh/known_hosts` → Mendaftarkan GitHub sebagai server yang aman dan dipercaya. Perintah ini secara otomatis memindai identitas keamanan (kunci publik) dari `github.com`, lalu menyimpannya ke dalam file daftar kontak terpercaya (`known_hosts`) supaya koneksi selanjutnya tidak ditolak

```
root@myserver:/home/delops# mkdir -p ~/.ssh
root@myserver:/home/delops# ssh-keyscan github.com >> ~/.ssh/known_hosts
# github.com:22 SSH-2.0-3992d52
# github.com:22 SSH-2.0-3992d52
# github.com:22 SSH-2.0-3992d52
# github.com:22 SSH-2.0-3992d52
# github.com:22 SSH-2.0-3992d52
root@myserver:/home/delops#
```

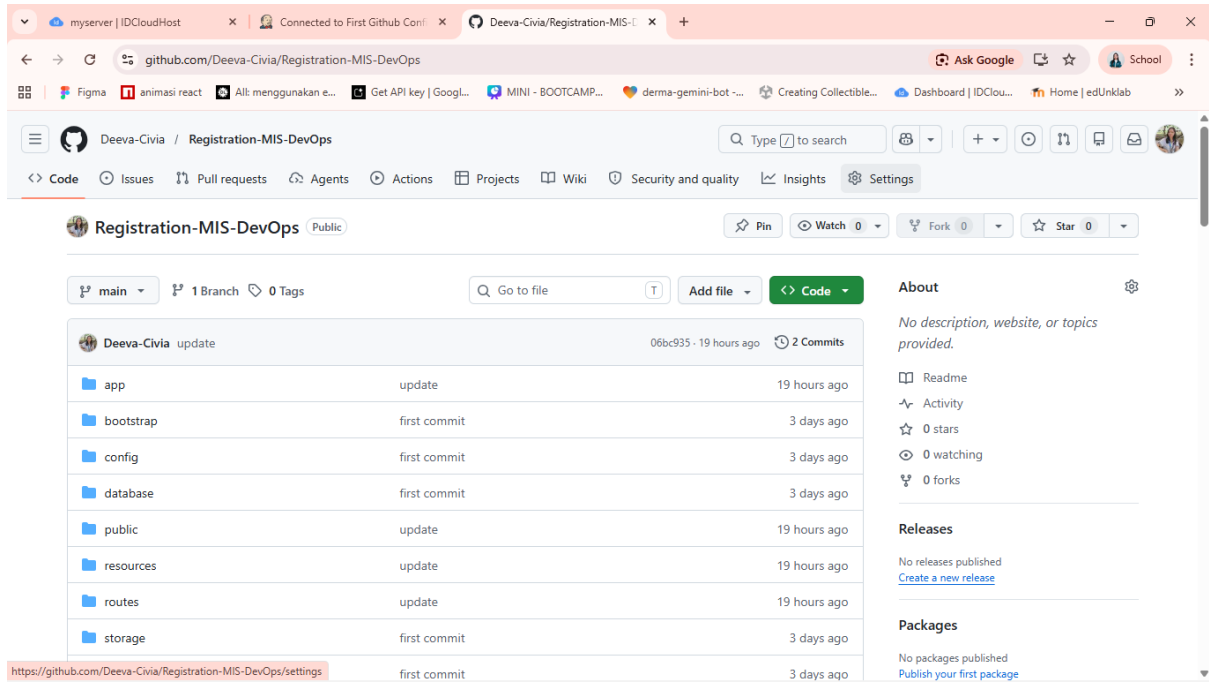
Masukkan nama branch yang ada di github yaitu “main”, lalu klik save.

The screenshot shows the Jenkins web interface in a browser. The page title is "Configure" under the "Connected to First Github" job. On the left, there's a sidebar with navigation links: General, Source Code Management (selected), Triggers, Environment, Build Steps, and Post-build Actions. The main content area is for configuring the Source Code Management plugin. It includes a "Repository URL" field with the value "git@github.com:Deeva-Civia/Registration-MIS-DevOps.git", a "Credentials" dropdown menu showing "Deeva-Civia (Deeva's github)", and an "Advanced" dropdown. Below these is a "+ Add Repository" button. The "Branches to build" section has a "Branch Specifier (blank for 'any')" field with the value "*/main". At the bottom, there are "Save" and "Apply" buttons. The "Save" button is highlighted in blue.

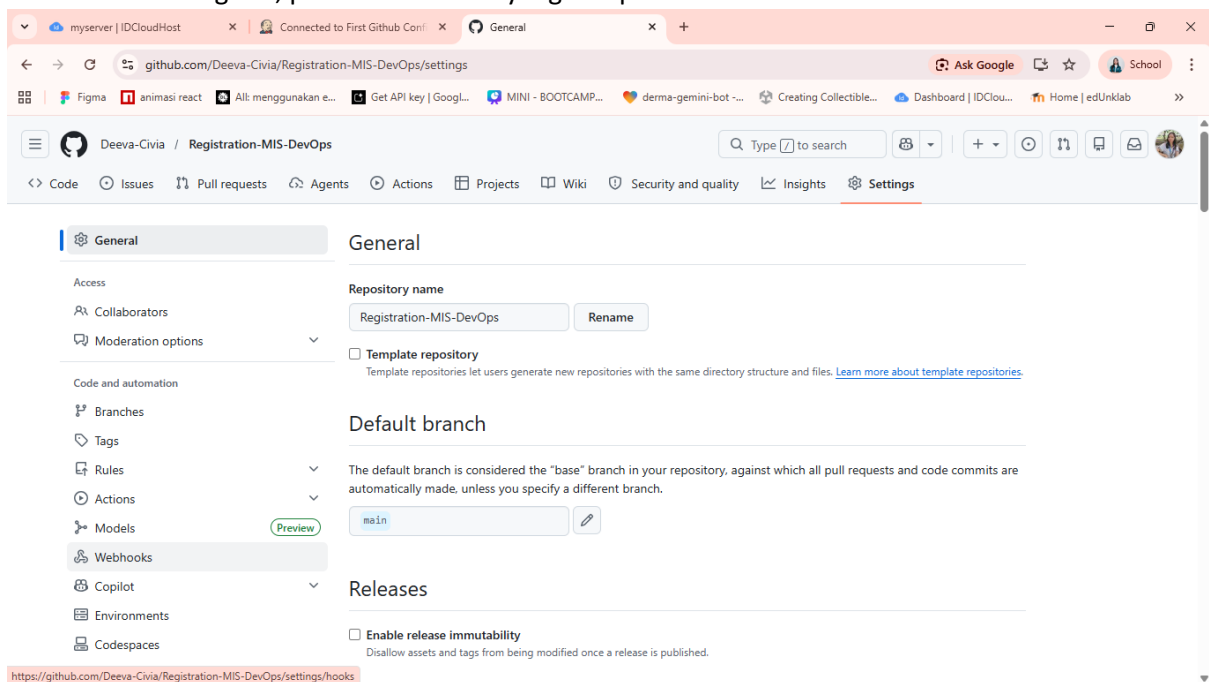
Selanjutnya kita akan membuat Webhook.

Webhook disini berperan sebagai alarm yang akan melakukan trigger atau memberi tahu Jenkins jika kita baru saja melakukan push dari local ke github. Hal ini agar jenkins tidak perlu melakukan pengecekan tiap berapa menit, melainkan hanya akan mengambil perubahan jika memang ada push terbaru.

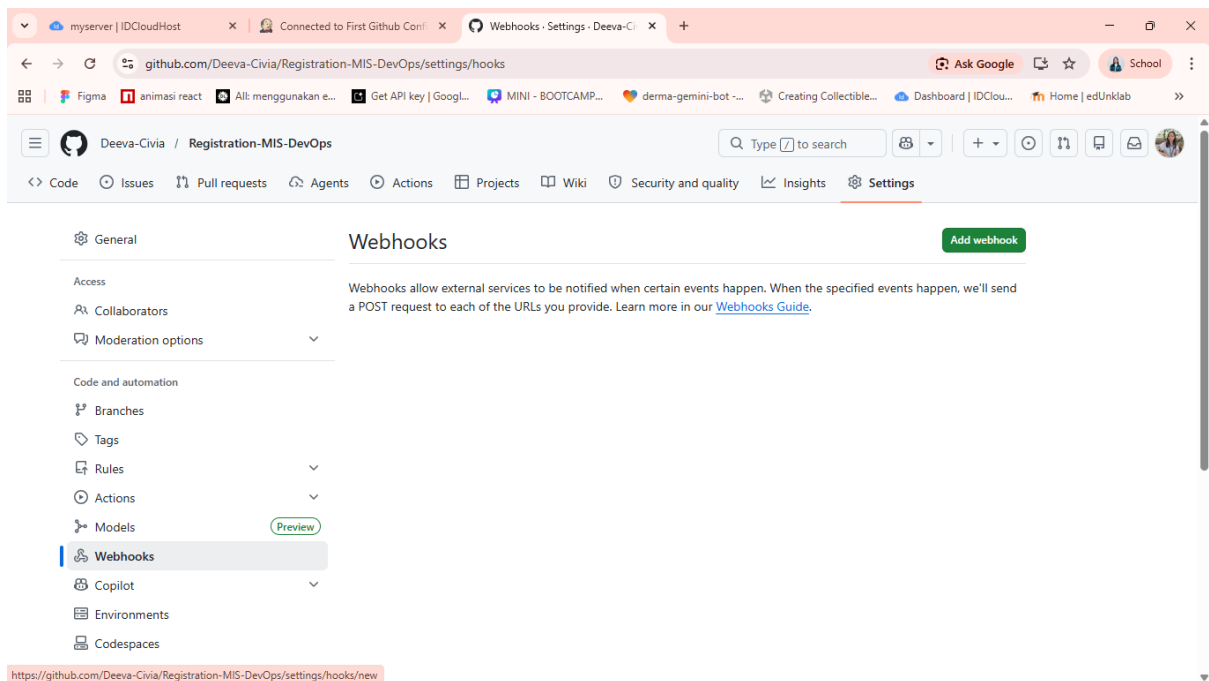
Buka Repository Github, dan pilih menu “Settings” yang ada di dalam repository



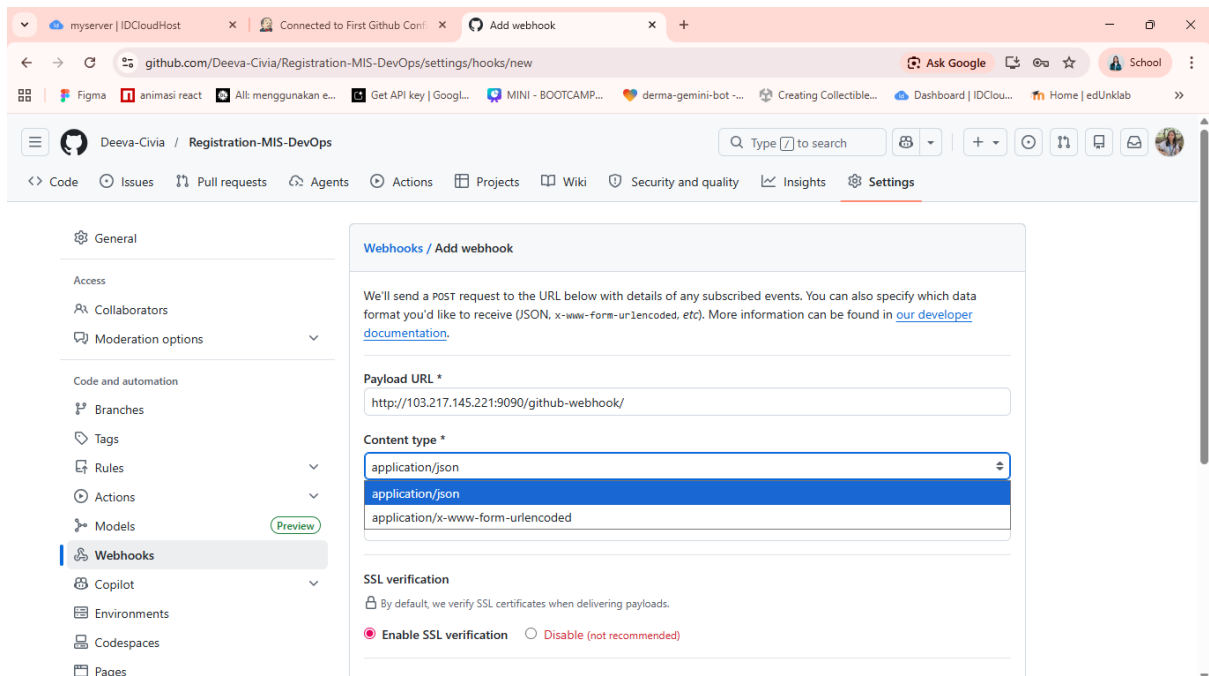
Pada menu settings ini, pilih “Webhooks” yang ada pada side bar:



Dalam menu Webhooks, klik “Add webhook”



Untuk bagian ini, pada Payload URL diisi dengan IP server dengan port Jenkins yaitu `http://<IP_SERVER>:9090/github-webhook/` dan untuk bagian Content Type, pilih “application/json”.



Dan pada bagian “Which events would you like to trigger this webhook?”, pilih “Just the push event”
Lalu untuk pengaturan lainnya dibiarkan default seperti itu, dan klik “Add webhook”

The screenshot shows the 'Add webhook' interface in a web browser. The left sidebar contains navigation links: Rules, Actions, Models (with a 'Preview' button), Webhooks (selected), Copilot, Environments, Codespaces, and Pages. Below these are sections for 'Security and quality' (Advanced Security, Deploy keys, Secrets and variables) and 'Integrations' (GitHub Apps, Email notifications). The main content area is titled 'Add webhook' and includes a dropdown menu set to 'application/json', a 'Secret' input field, and 'SSL verification' options (Enable SSL verification is selected). Under the heading 'Which events would you like to trigger this webhook?', the option 'Just the push event.' is selected. Other options are 'Send me everything.' and 'Let me select individual events.' The 'Active' checkbox is checked, with a note: 'We will deliver event details when this hook is triggered.' A green 'Add webhook' button is at the bottom.

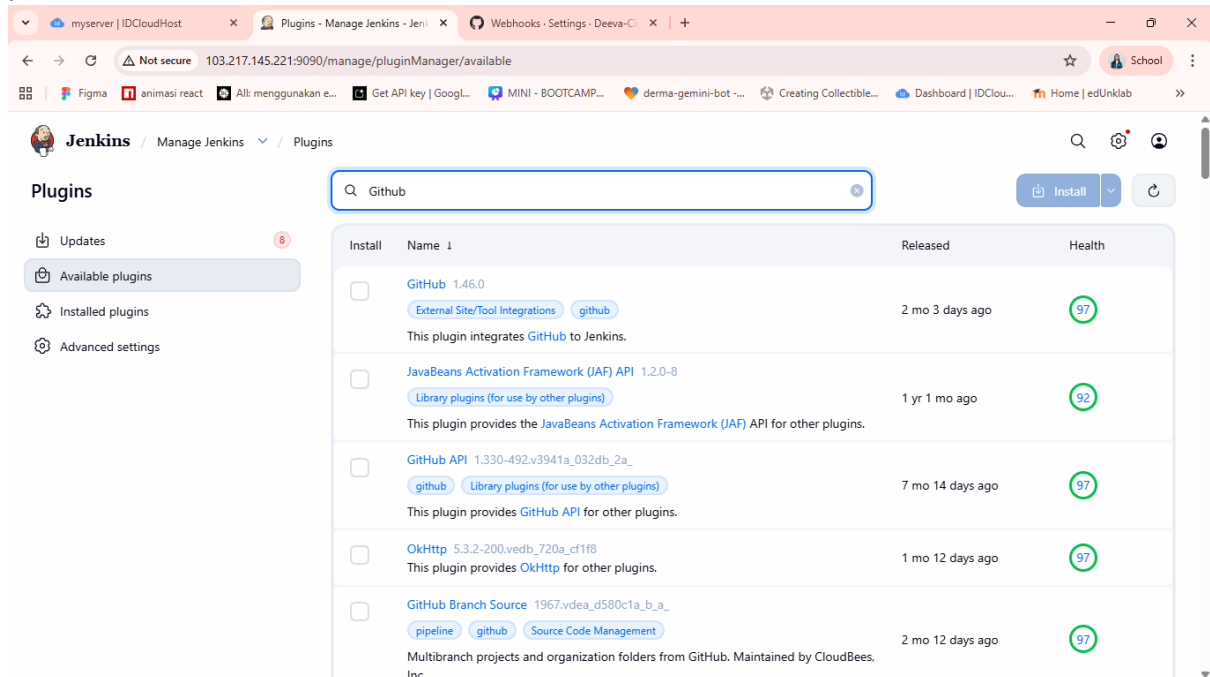
Tampilan setelah Webhook berhasil dibuat:

The screenshot shows the 'Webhooks' settings page after successful creation. A blue notification banner at the top states: 'Okay, that hook was successfully created. We sent a ping payload to test it out! Read more about it at https://docs.github.com/webhooks/#ping-event.' The left sidebar is identical to the previous screenshot, with 'Webhooks' selected. The main content area is titled 'Webhooks' and includes an 'Add webhook' button. Below the title, a description explains that webhooks allow external services to be notified when certain events happen. A single webhook is listed with the URL 'http://103.217.145.221:9090/github...' (push) and buttons for 'Edit' and 'Delete'. A note below the URL states: 'This hook has never been triggered.'

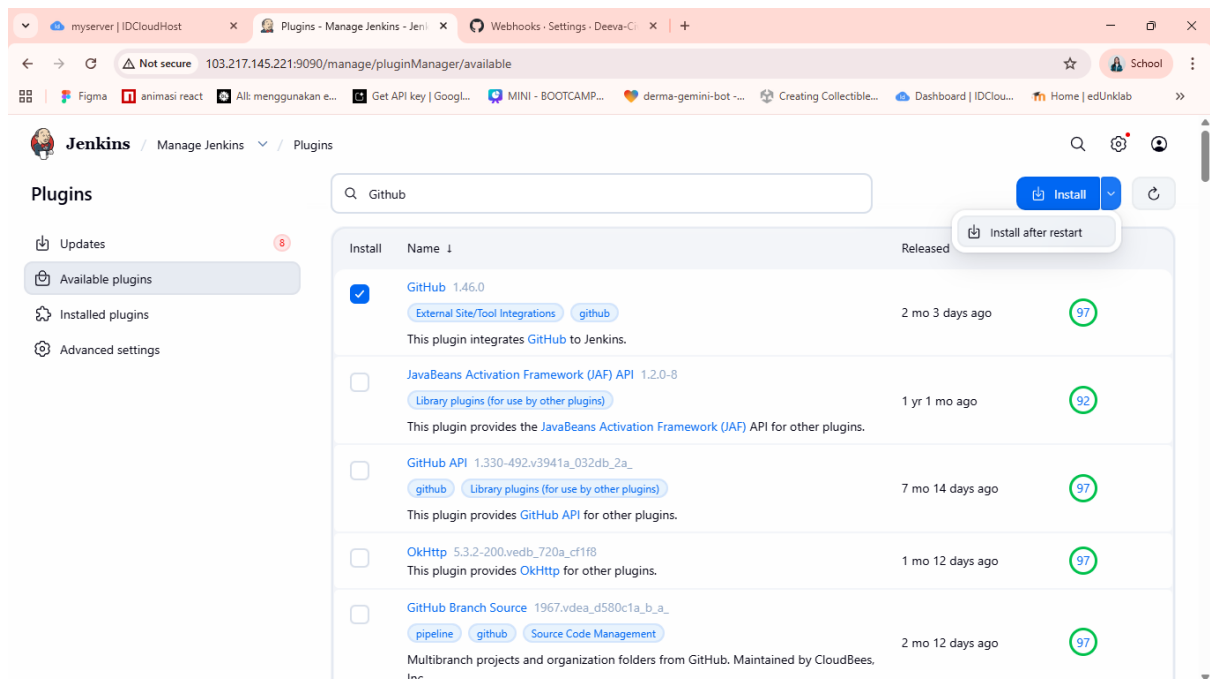
Sekarang kita beri tahu Jenkins untuk selalu mendengarkan alarm dari Github, dengan cara mengatur trigger pada job yang telah kita buat tadi.

Namun sebelum itu, kita harus melakukan install plugin khusus untuk Github.

Caranya, kembali ke menu Pluggins, dan masuk ke “Available pluggins” lalu cari “Github” pada kolom pencarian.



Centang lalu klik “Install after restart”



myserver | IDCloudHost x Update Center - Manage Jenkins x Webhooks - Settings - Deeva-Ci x +

Not secure 103.217.145.221:9090/manage/pluginManager/updates/ School

Figma animasi react Alt: menggunakan e... Get API key | Googl... MINI - BOOTCAMP... derma-gemini-bot ... Creating Collectible... Dashboard | IDClo... Home | edUnklab

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates 8
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

Download progress

Preparation

- Checking internet connectivity
- Checking update center connectivity
- Success

OkHttp	Downloaded Successfully. Will be activated during the next boot
JavaBeans Activation Framework (JAF) API	Downloaded Successfully. Will be activated during the next boot
JAXB	Downloaded Successfully. Will be activated during the next boot
JSON Api	Downloaded Successfully. Will be activated during the next boot
SnakeYAML API	Downloaded Successfully. Will be activated during the next boot
Jakarta XML Binding API	Downloaded Successfully. Will be activated during the next boot
Woodstox Core API	Downloaded Successfully. Will be activated during the next boot
Jackson 2 API	Downloaded Successfully. Will be activated during the next boot
GitHub API	Downloaded Successfully. Will be activated during the next boot
Ionicons API	Downloaded Successfully. Will be activated during the next boot
ASM API	Downloaded Successfully. Will be activated during the next boot
SCM API	Downloaded Successfully. Will be activated during the next boot
Pipeline: SCM Step	Downloaded Successfully. Will be activated during the next boot
Credentials	Downloaded Successfully. Will be activated during the next boot
Gson API	Downloaded Successfully. Will be activated during the next boot

Centang bagian “Restart Jenkins when installation is complete and no jobs are running”

myserver | IDCloudHost x Update Center - Manage Jenkins x Webhooks - Settings - Deeva-Ci x +

Not secure 103.217.145.221:9090/manage/pluginManager/updates/ School

Figma animasi react Alt: menggunakan e... Get API key | Googl... MINI - BOOTCAMP... derma-gemini-bot ... Creating Collectible... Dashboard | IDClo... Home | edUnklab

Jenkins / Manage Jenkins / Plugins

Plugins

- Updates 8
- Available plugins
- Installed plugins
- Advanced settings
- Download progress

GitHub API	Downloaded Successfully. Will be activated during the next boot
Ionicons API	Downloaded Successfully. Will be activated during the next boot
ASM API	Downloaded Successfully. Will be activated during the next boot
SCM API	Downloaded Successfully. Will be activated during the next boot
Pipeline: SCM Step	Downloaded Successfully. Will be activated during the next boot
Credentials	Downloaded Successfully. Will be activated during the next boot
Gson API	Downloaded Successfully. Will be activated during the next boot
Git client	Downloaded Successfully. Will be activated during the next boot
Git	Downloaded Successfully. Will be activated during the next boot
JSON Path API	Downloaded Successfully. Will be activated during the next boot
Token Macro	Downloaded Successfully. Will be activated during the next boot
GitHub	Downloaded Successfully. Will be activated during the next boot

→ [Go back to the top page](#)
(you can start using the installed plugins right away)

→ ☐ Restart Jenkins when installation is complete and no jobs are running

REST API Jenkins 2.555.1

Setelah itu, Jenkins akan melakukan restart.

Selesai restart, kita login kembali, dan masuk ke menu Job yang telah kita buat tadi.

“Connected to First Github”

myserver | IDCloudHost

Dashboard - Jenkins

Webhooks - Settings - Deeva-Ci

Not secure 103.217.145.221:9090

School

Figma

animasi react

All: menggunakan e...

Get API key | Googl...

MINI - BOOTCAMP...

derma-gemini-bot ...

Creating Collectible...

Dashboard | IDClo...

Home | edUnklab

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

+

S	W	Name	Last Success	Last Failure	Last Duration
🕒	☀️	Backup Database	16 days #4	N/A	0.2 sec
🕒	☀️	BMKG	1 mo 0 days #3	N/A	0.69 sec
🕒	☀️	Connected to First Github	N/A	N/A	N/A
🕒	☀️	Simple Automated Message	1 mo 19 days #5	N/A	22 ms

Icon: S M L

REST API

Jenkins 2.555.1

Klik “Configure”

myserver | IDCloudHost

Connected to First Github - Jeni

Webhooks - Settings - Deeva-Ci

Not secure 103.217.145.221:9090/job/Connected%20to%20First%20Github/

School

Figma

animasi react

All: menggunakan e...

Get API key | Googl...

MINI - BOOTCAMP...

derma-gemini-bot ...

Creating Collectible...

Dashboard | IDClo...

Home | edUnklab

Jenkins

/ Connected to First Github

Status

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Builds

No builds

Connected to First Github

Connect ke github repository

Permalinks

Edit description

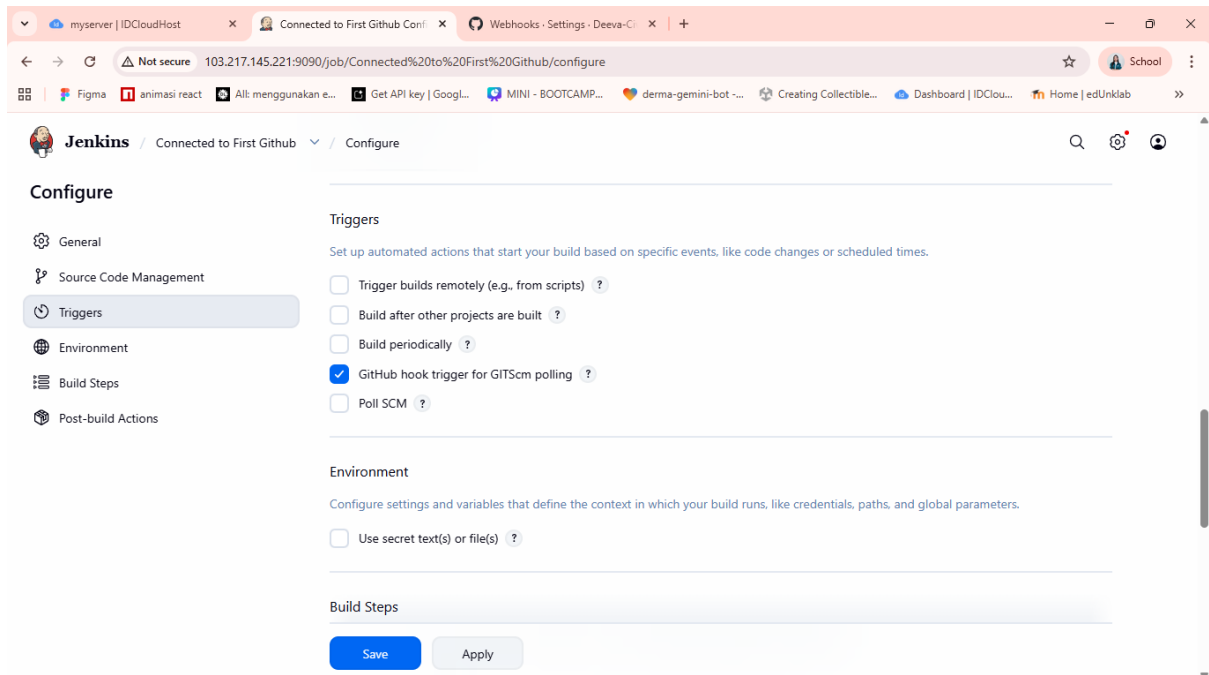
REST API

Jenkins 2.555.1

103.217.145.221:9090/job/Connected to First Github/configure

Dalam menu Configure, pilih menu “Triggers” yang ada pada side bar

Centang bagian “GitHub hook trigger for GITScm polling”, lalu klik save



Selanjutnya saat Jenkins mendapat code terbaru dari GitHub, code itu masuk ke dalam Workspace internal Jenkins. Kita harus menyuruh Jenkins memindahkan code tersebut ke directory project website kita yaitu `/var/www/html/registration-mis`

Caranya, masuk lagi ke menu Configure dan pilih menu “Build Step” pada side bar, dan pilih “Execute shell”.

Dalam field Execute shell, masukkan code dibawah ini:

`rsync -avz --exclude '.git' --exclude '.env' ./ /var/www/html/registration-mis/` → untuk sinkronisasi file dari internal Jenkins (workspace) ke direktori web server tujuan.

Detail Parameter:

- `rsync`: Perintah transfer file yang lebih cepat karena hanya mengirimkan bagian file yang berubah
- `-a` (archive): Menjaga struktur direktori, hak akses (permissions), dan symlink file tetap sama seperti aslinya
- `-v` (verbose): Menampilkan proses transfer file di log Jenkins
- `-z` (compress): Compress data selama proses transfer supaya lebih ringan dan cepat
- `--exclude '.git'`: Mencegah folder riwayat GitHub ikut tersalin ke web server publik
- `--exclude '.env'`: Mencegah file env ikut tertimpa.
- `./`: Direktori sumber yaitu workspace Jenkins saat ini
- `/var/www/html/registration-mis/`: Direktori tujuan tempat project website

`cd /var/www/html/registration-mis/` → untuk masuk ke dalam folder project Laravel yang ada di server.

`composer install --optimize-autoloader --no-dev` → Mengelola dan menginstal library PHP yang dibutuhkan Laravel.

`php artisan migrate --force` → Menjalankan file migrasi PHP yang baru ditambahkan ke repositori untuk memperbarui skema database di server. Menggunakan Force agar langsung mengeksekusi migrasi.

`chown -R www-data:www-data /var/www/html/registration-mis` → Mengubah kepemilikan folder dan seluruh isinya kepada user `www-data` dan group `www-data`.

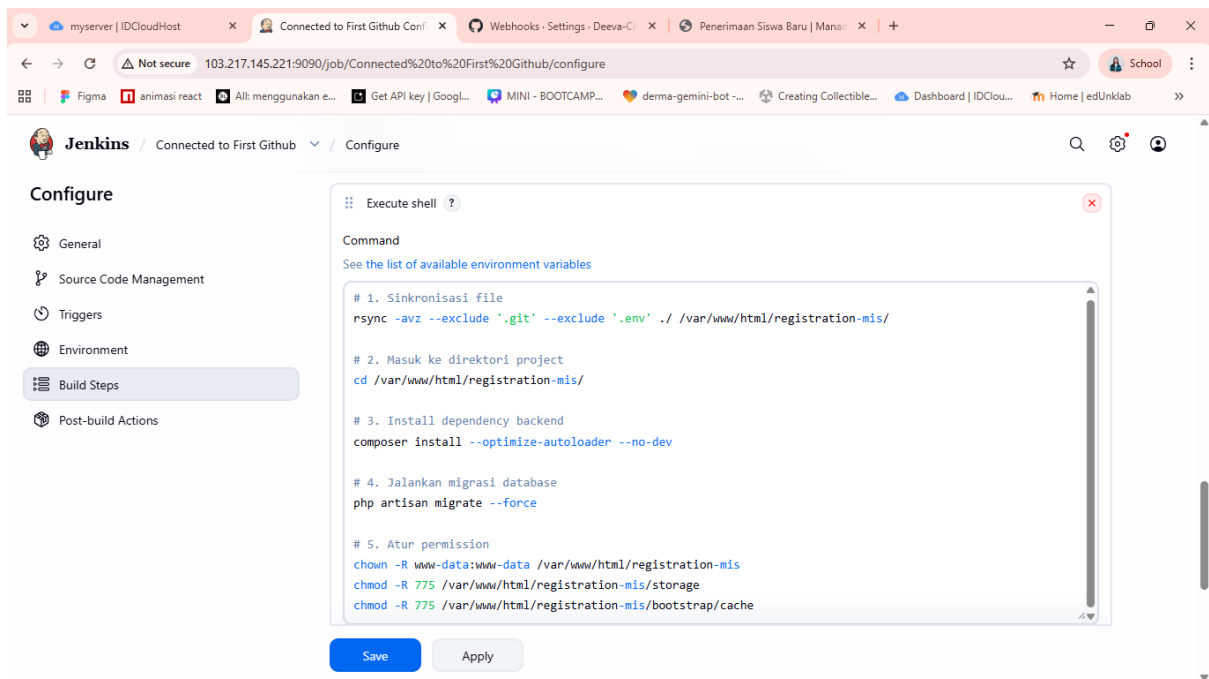
`www-data` adalah user standar Linux untuk web server, sehingga web server memiliki hak untuk membaca project tersebut.

`chmod -R 775 /var/www/html/registration-mis/storage` → Mengubah hak akses folder storage. Angka 775 memberikan hak baca, tulis, dan eksekusi kepada pemilik dan grup

`chmod -R 775 /var/www/html/registration-mis/bootstrap/cache` → Mengubah hak akses folder cache. Angka 775 memberikan hak baca, tulis, dan eksekusi kepada pemilik dan grup

Semua code ini mulai dari `composer install` sampai pada memberikan hak akses untuk folder cache, dikarenakan saya menggunakan Laravel Full Stack untuk website saya.

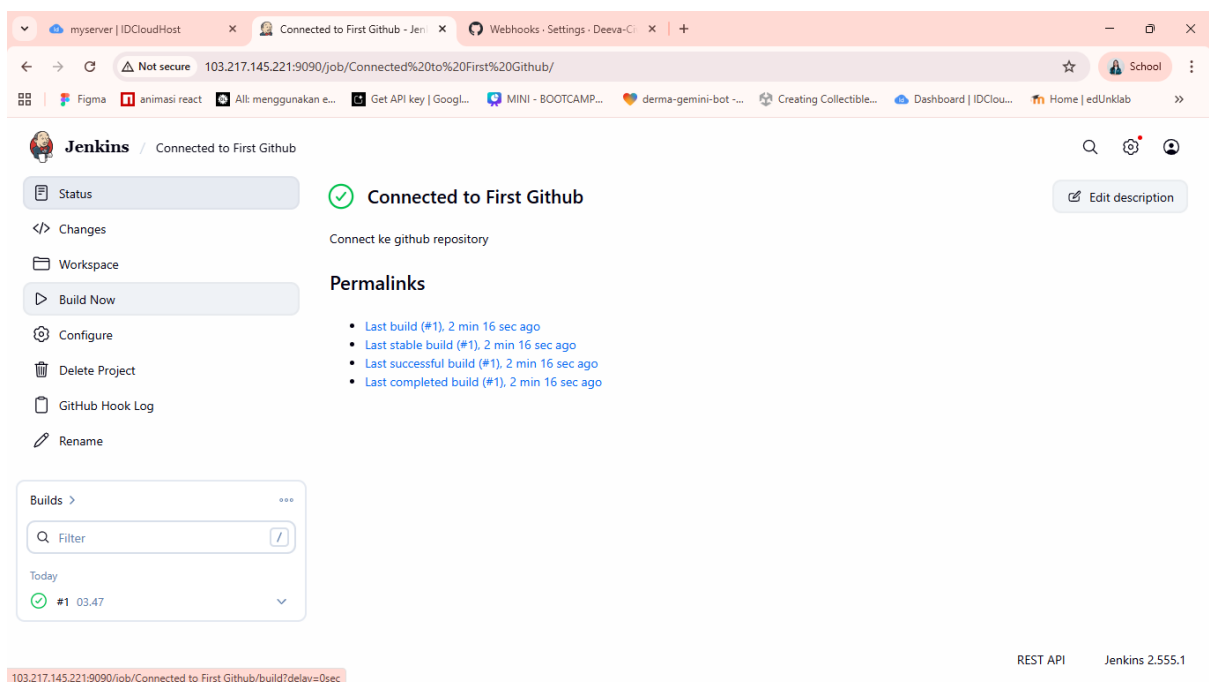
Setelah mengisi field Execute shell, klik save



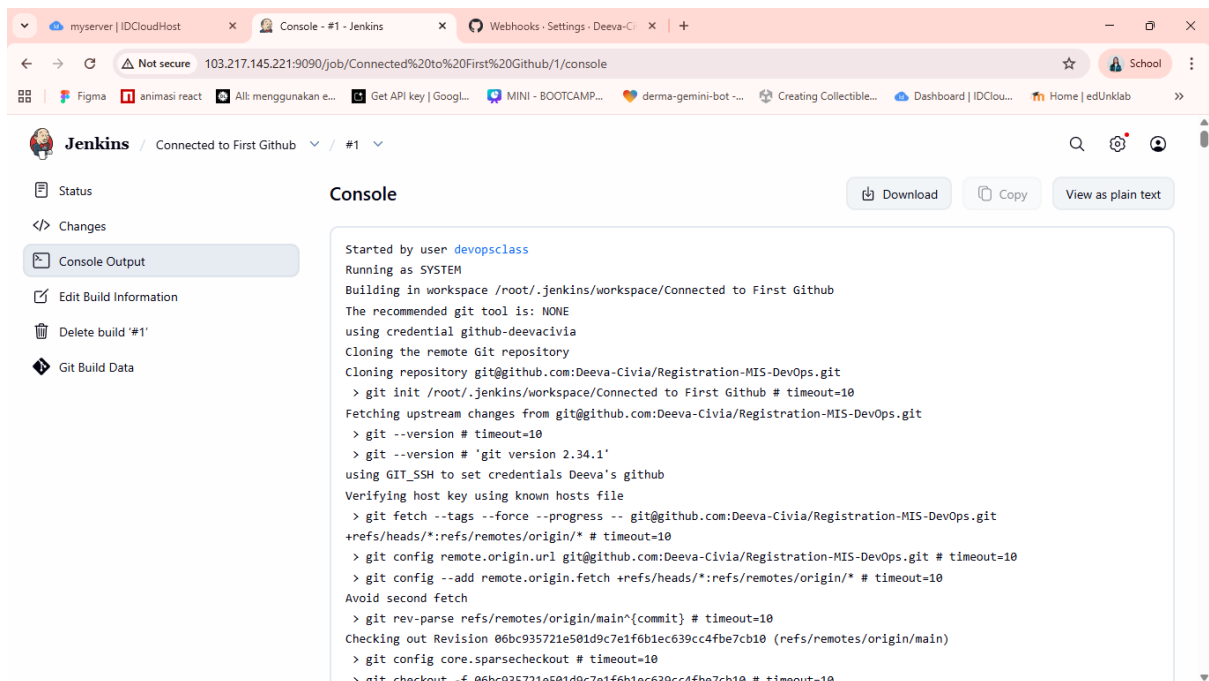
Setelah selesai melakukan konfigurasi itu, klik “Build Now” pada side bar, untuk inisiasi awal yaitu membuat workspace jenkins dan clone project.

Dan jika telah selesai, kita dapat melihat pada kolom Build terdapat “#1” berwarna hijau yang berarti proses berhasil dilakukan.

Klik “#1” tersebut, lalu buka Console Output untuk melihat detail prosesnya.



Ini isi dari console output, scroll sampai bawah dan kita akan melihat pesan Success

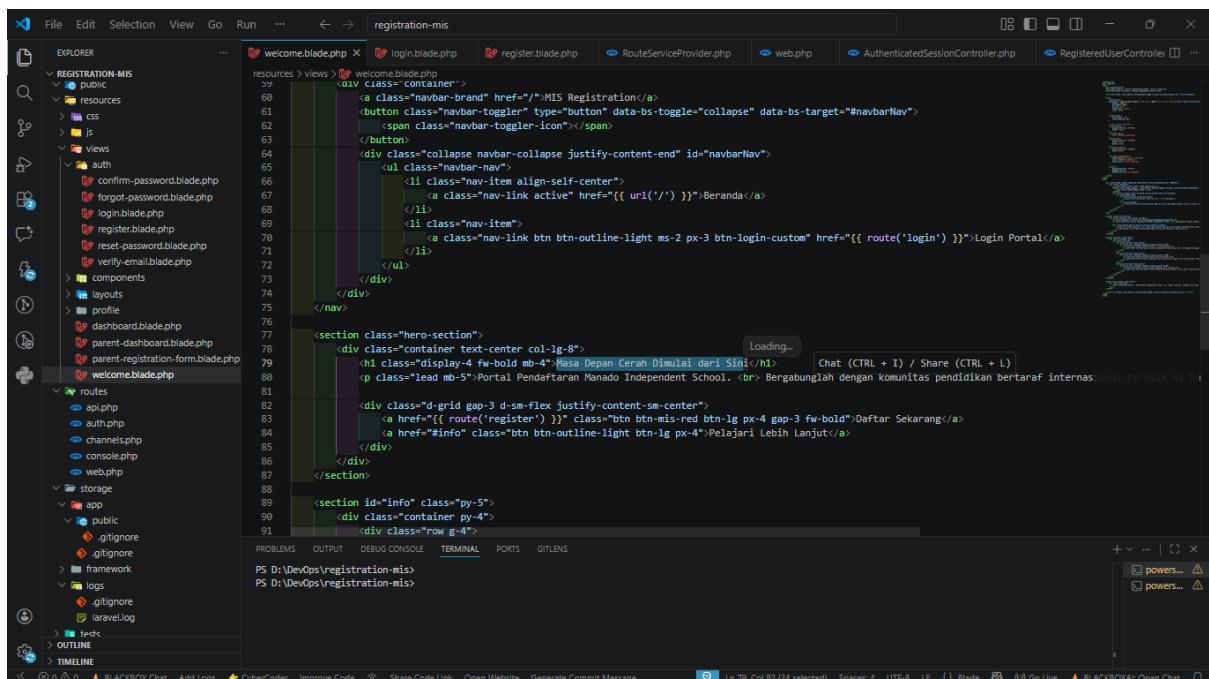


```
Started by user devopsclass
Running as SYSTEM
Building in workspace /root/.jenkins/workspace/Connected to First Github
The recommended git tool is: NONE
using credential github-deevacivia
Cloning the remote Git repository
Cloning repository git@github.com:Deeva-Civia/Registration-MIS-DevOps.git
> git init /root/.jenkins/workspace/Connected to First Github # timeout=10
Fetching upstream changes from git@github.com:Deeva-Civia/Registration-MIS-DevOps.git
> git --version # timeout=10
> git --version # 'git version 2.34.1'
using GIT_SSH to set credentials Deeva's github
Verifying host key using known hosts file
> git fetch --tags --force --progress -- git@github.com:Deeva-Civia/Registration-MIS-DevOps.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url git@github.com:Deeva-Civia/Registration-MIS-DevOps.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 06bc935721e501d9c7e1f6b1ec639cc4fbc7cb10 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 06bc935721e501d9c7e1f6b1ec639cc4fbc7cb10 # timeout=10
```

6. Pengujian CI/CD Real-time

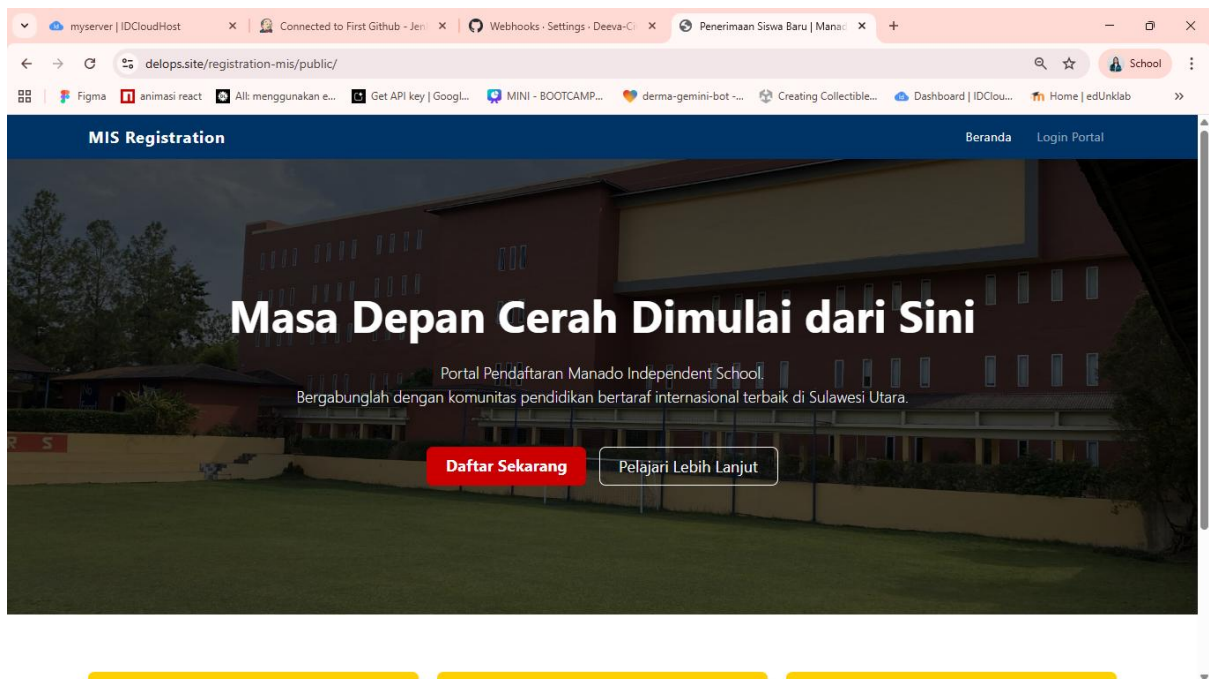
Untuk mengetes apakah CI/CD yang telah dibuat benar berhasil, buka folder project di local melalui VS code.

ini isi dari code Landing Page website saya.



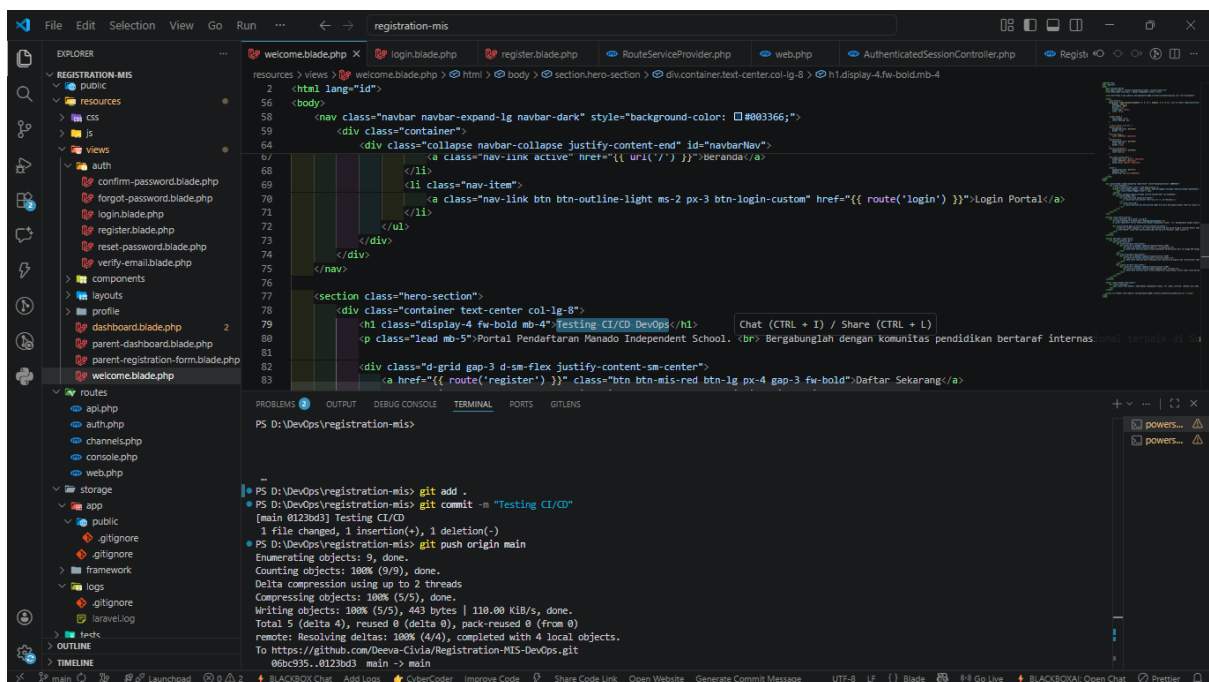
```
resources > views > welcome.blade.php
39
40 <div class="container">
41     <a class="navbar-brand" href="/MIS Registration/">
42     <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarNav">
43     <span class="navbar-toggler-icon"></span>
44     </button>
45     <div class="collapse navbar-collapse justify-content-end" id="navbarNav">
46     <ul class="navbar-nav">
47     <li class="nav-item align-self-center">
48     <a class="nav-link active" href="{{ url('/') }}">Beranda</a>
49     </li>
50     <li class="nav-item">
51     <a class="nav-link btn btn-outline-light ms-2 px-3 btn-login-custom" href="{{ route('login') }}">Login Portal</a>
52     </li>
53     </ul>
54     </div>
55     </div>
56     </nav>
57 </div>
58
59 <section class="hero-section">
60 <div class="container text-center col-lg-8">
61 <h1 class="display-4 fw-bold mb-4">Masa Depan Cerah Dimulai dari Sini</h1>
62 <p class="lead mb-5">Portal Pendaftaran Manado Independent School. <br> Bergabunglah dengan komunitas pendidikan bertaraf internasional yang terdapat di SI
63 </p>
64 <div class="d-grid gap-3 d-sm-flex justify-content-center">
65 <a href="{{ route('register') }}" class="btn btn-mis-red btn-lg px-4 gap-3 fw-bold">Daftar Sekarang</a>
66 <a href="#info" class="btn btn-outline-light btn-lg px-4">Pelajari Lebih Lanjut</a>
67 </div>
68 </div>
69 </section>
70
71 <section id="info" class="py-5">
72 <div class="container py-4">
73 <div class="row g-4">
74
75 </div>
76 </div>
77 </section>
78
79 </div>
80 </div>
81 </div>
82 </div>
83 </div>
84 </div>
85 </div>
86 </div>
87 </div>
88 </div>
89 </div>
90 </div>
91 </div>
```

Dan ini adalah tampilan Landing Page sebelum saya melakukan perubahan

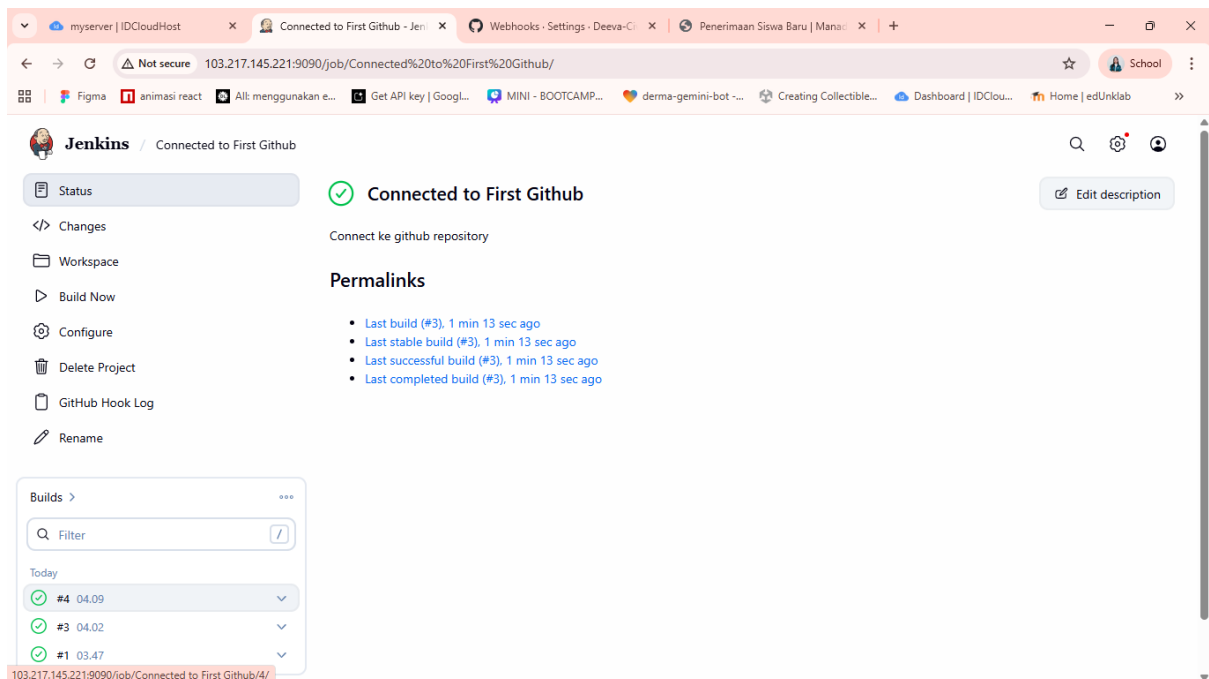


Sekarang saya akan mengubah Text “Masa Depan Cerah Dimulai dari Sini” menjadi “Testing CI/CD DevOps” melalui VS code. Lalu melakukan commit dan push ke github.

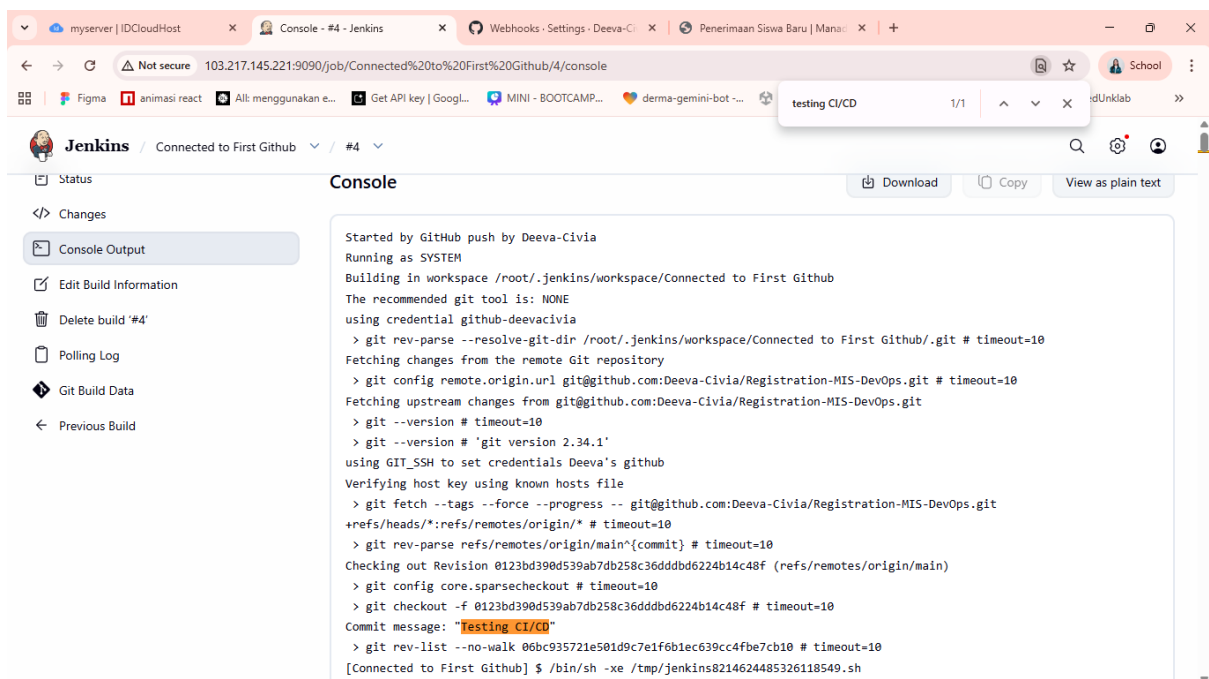
Disini saya telah selesai push.



Cek Jenkins, pada job di jenkins telah ada build terbaru “#4” dan kita akan mengecek console outputnya



Ini adalah console output dari build terbaru, dan dapat terlihat disini bahwa jenkins berhasil mengambil commit yang telah saya lakukan tadi.



Cek lagi tampilan Landing Page Website, disini tulisannya telah berhasil berubah.

Yang berarti code yang telah kita push dari local telah berhasil diambil oleh Jenkins secara otomatis.

